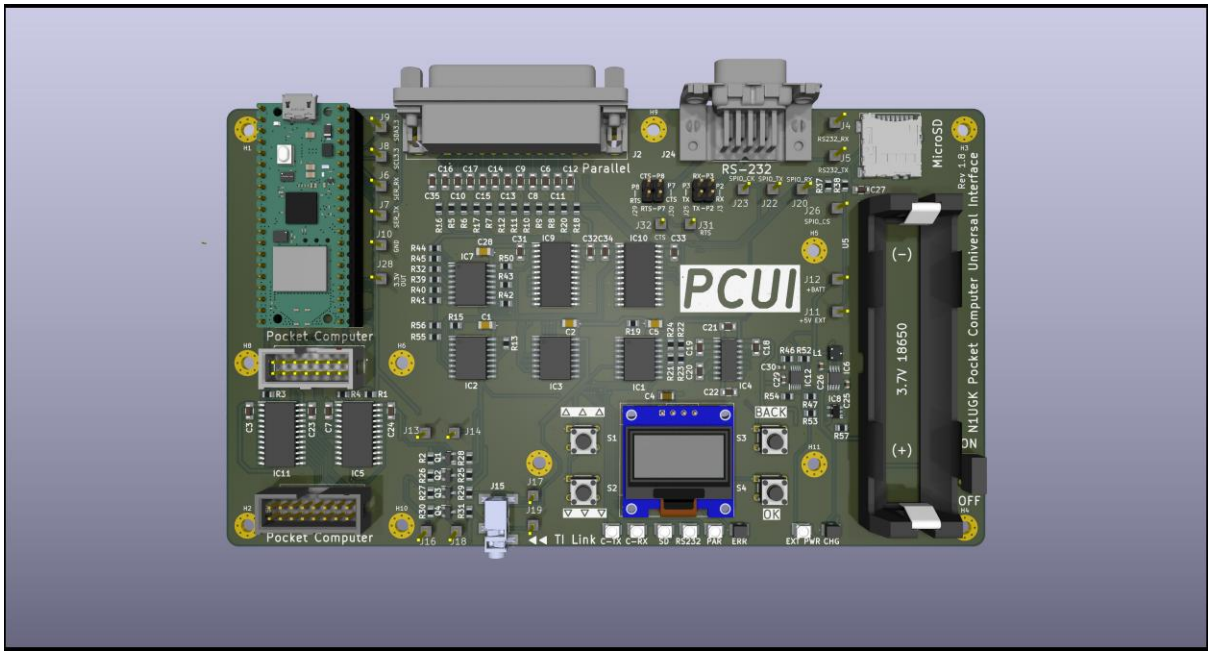


Pocket Computer Universal Interface (PCUI)

Hardware Version 1.8

Document Version 1.4

October 10, 2026

Assembly Instructions

I. Introduction

Even though these instructions are intended to be as detailed as possible, some experience is absolutely required. Not only is assembly required, surface mount device (SMD) is utilized throughout the design and sourcing of components for the PCUI.

Unfortunately, through-hole versions of components were not always available for some integrated circuits needed for this project. Therefore, experience soldering SMD components will be required.

Below are some of the tools needed for assembling the PCUI:

1. **Soldering Iron**

   A good quality temperature stabilized, and temperature adjustable soldering iron is required. Further, due to SMD components used in this design, an SMD soldering iron tip is required. For example, I have had good results using this soldering tip for SMD: Hakko T18-BR02 - Bent Conical, 0.2mm Radius/30°

   If you are prone to making a few mistakes, I recommend solder wick for removing solder.

   Be sure to use a quality solder you are familiar with – whether it be leaded or lead-free, know the best temperature for your type of solder and soldering tip.

2. **Magnification Glasses**

   Unless you have super-human eyesight, magnified (and optionally lighted) glasses will be needed, especially for the smallest of the SMD components.

3. **Test Equipment**

   A DMM is required for measuring voltage, resistance, and continuity. A bench power supply is helpful for testing the power supply section. The supply should be capable of current limiting and a 3.7V DC output.

B. Safety

This kit operates on 5 volts and 3.3 volts and is considered "low voltage". However, safety procedures should be followed to prevent damage to USB ports, batteries, and other components. Lithium-ion batteries can discharge quickly and some are low quality and lack safety measures found in modern battery management systems (BMS). Not all lithium-ion batteries are created equal nor are they all protected from short circuits. Use caution when working with lithium-ion batteries and USB ports.

Regarding lithium-ion batteries, the PCUI is not shipped with a lithium-ion battery due to shipping regulations and safety. A battery is recommended but is not required for the PCUI. If not using a battery, then a USB power source would be required such as a power bank or computer.

C. Patience

Follow these instructions carefully and do not rush. Fully understand the instructions before proceeding. This will help prevent mistakes. Perhaps sharpen your soldering skills by practicing on spare boards and parts before jumping into this kit or a particular step in this build.

D. Hardware Versioning

Please be sure you are reading the most recent document version for the specific hardware version you have. Because the PCUI is distributed either assembled or as a kit, and there are several variants of the hardware that have been distributed, this document is updated accordingly.

The hardware version is provided as silkscreen lettering on the PCUI board. It will be marked similar to "Rev 1.8" as an example.

E. Order of Assembly

Do try and follow the order of assembly outlined in this guide. The reason for doing so is to be sure you can test along the way, reducing the risk of damage to components should a mistake be made.

II. Inventory of Parts

A. Kit Variations

The PCUI kit you purchased may not have all the interface packs outlined in this guide. Interface packs might be different for each model of pocket computer or calculator you wish to use with the PCUI.

B. Identifying Components

Many components themselves are not labelled or are not necessarily easy to identify. It is recommended to leave components in their packaging or with their labelling until they are needed.

Please note that any pictures included in these assembly instructions are for example only. The actual components you receive might have variations in color or appearance, as substitutions may be made as needed due to parts availability and pricing.

It was intended that all PCUI components be readily available from multiple sources. If you inadvertently damage a component, chances are you can purchase a replacement yourself if you wish, or from the same source as you received the PCUI kit from.

C. Parts and Components

1. Raspberry Pi Pico 2 W

Your PCUI kit will contain a variant of the Raspberry Pi Pico 2 W. It has been pre-flashed with the bootloader and the firmware available at the time it was flashed. The firmware will most likely need updating and can be done by connecting the PCUI to a computer. Instructions for updating the firmware are included in the Owners Manual.

2. PCUI Main PCB

The PCUI is comprised of a single main PCB and one or more pocket computer interface pack PCBs (if applicable). The main PCB will be identified with a hardware version number – be sure the version number on your kit matches the version number of this guide.

3. Pocket Computer Interface Packs

Your PCUI kit might have one or more pocket computer interface packs. These packs contain a small PCB, a connector, and some have additional components. An interface pack is designed for a particular line of pocket computers that share a specific bus interface or pinout.

4. Lithium Ion Battery

The PCUI uses an 18650 lithium-ion battery for portable use. The battery itself is not included due to shipping regulations. 18650 batteries can be purchased from many online retailers such as Amazon and are very inexpensive. The PCUI currently does not have a charging circuit for the battery, and therefore the battery should be charged using an external charger.

Note that a battery is not required for PCUI operation. If a battery is not used, then a USB based power source would be required.

5. Test Headers

The PCUI includes test header pin locations on the PCB which are optional. These would be useful for debugging or troubleshooting. They are convenient for attaching test leads and can be added later if you need them.

III. Getting Started

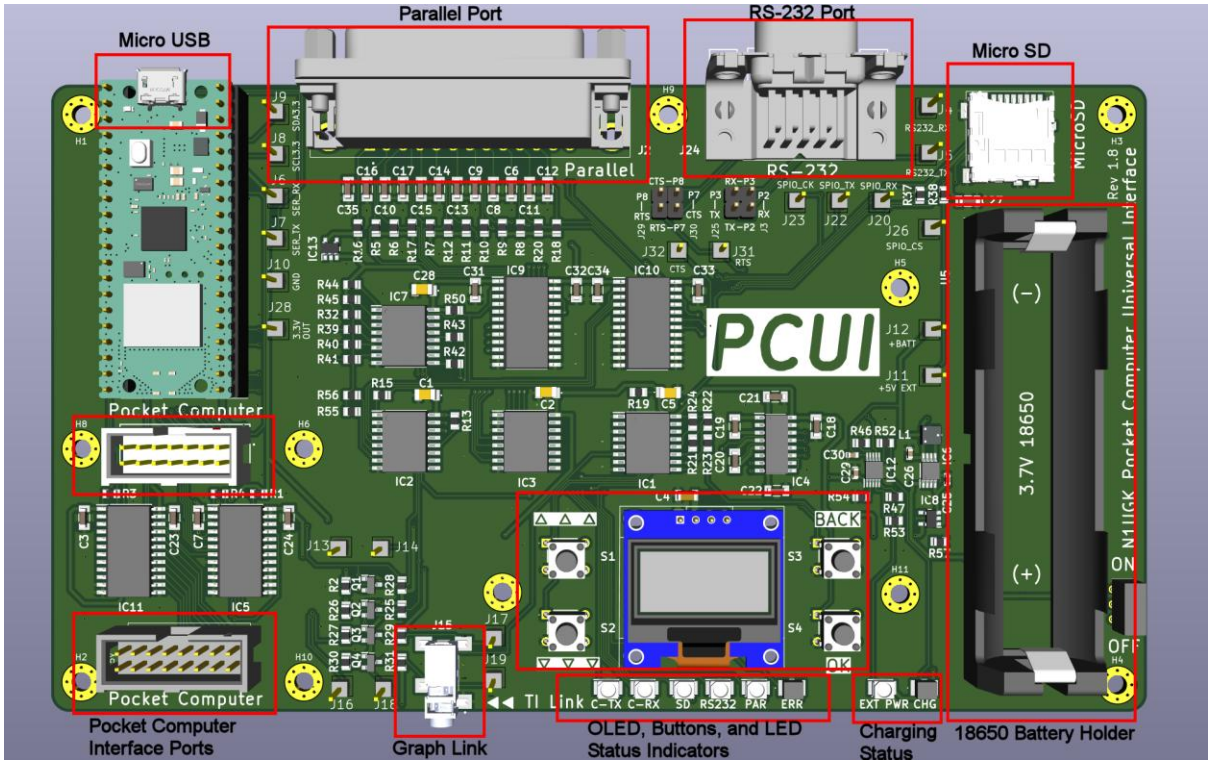
A. Unpacking and Inspection

When you first receive your PCUI kit, make sure all components are intact, not damaged, and avoid removing components from their packaging or removing labels until those components are needed. Many components are not individually labelled.

1. Inspect the packaging for any signs of damage during shipping.
2. Check that all components are present, based on the particular kit options you selected.

B. PCUI Main Board

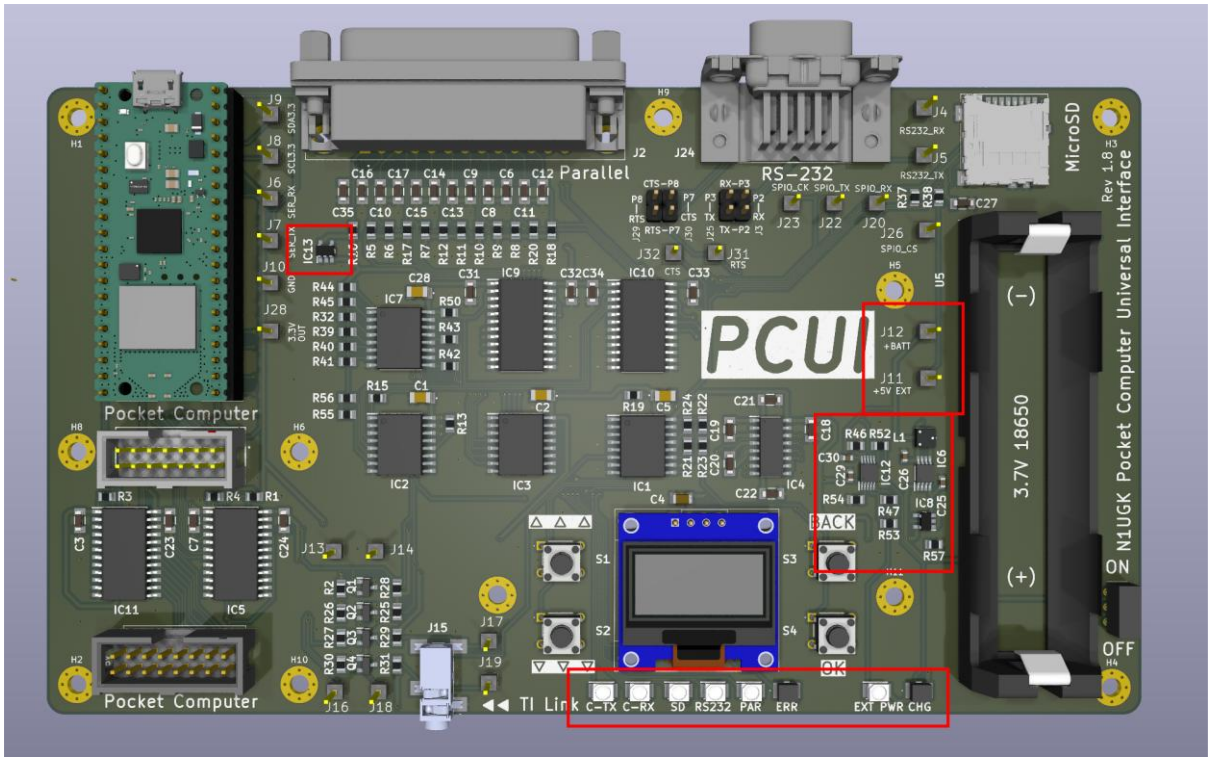
The fully assembled PCUI main board is shown below. Assembly instructions for each section if the main board is described in the next sections.


Micro USB
Parallel Port
RS-232 Port
Micro SD
Pocket Computer
Pocket Computer
Pocket Computer Interface Ports
Graph Link
OLED, Buttons, and LED Status Indicators
Charging Status
18650 Battery Holder
PCUI
Parallel
RS-232
MicroSD
3.7V 18650
(−)
(+)
BACK
TI Link
C-TX C-RX SD RS232 PAR ERR
EXT PWR CHG

IV. PCUI Main Board Assembly Instructions

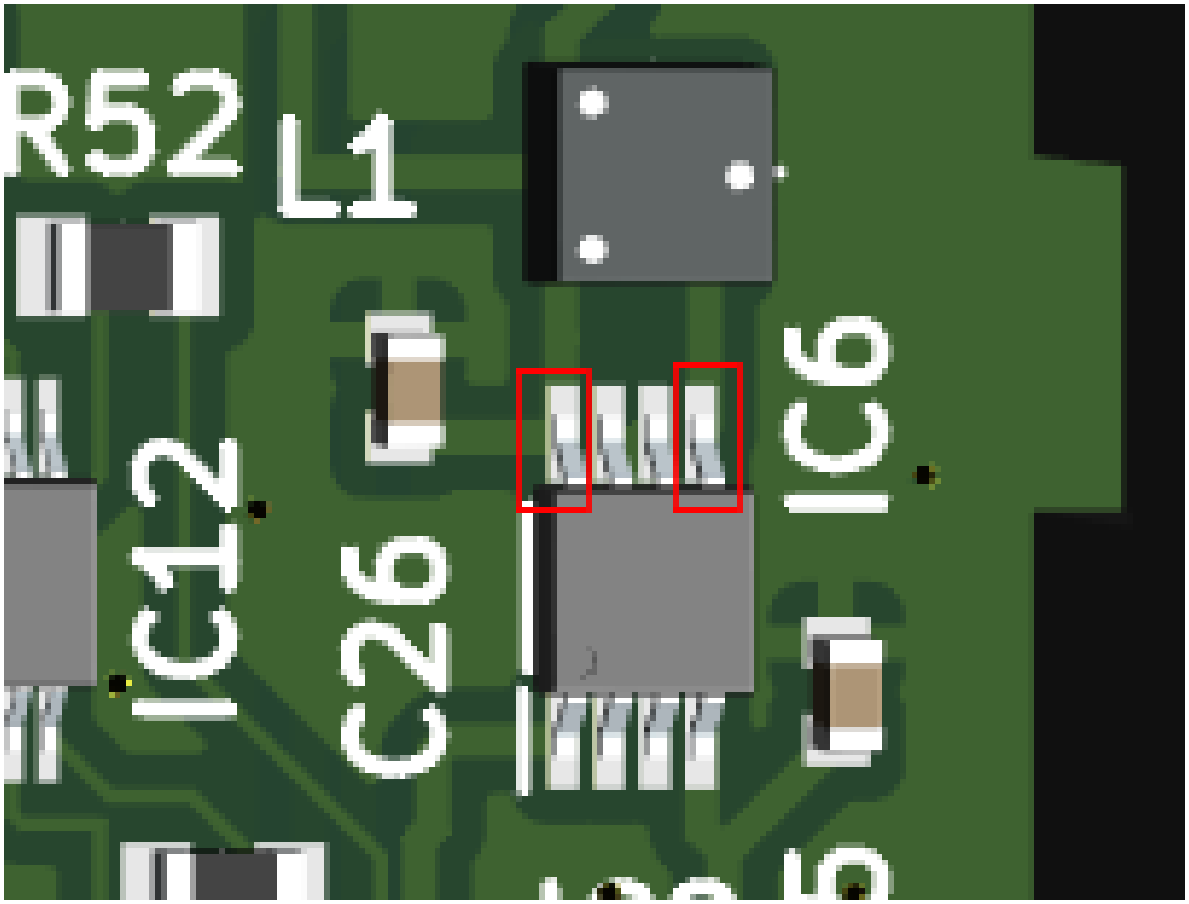
A. Power Supply Section

1. Assembly

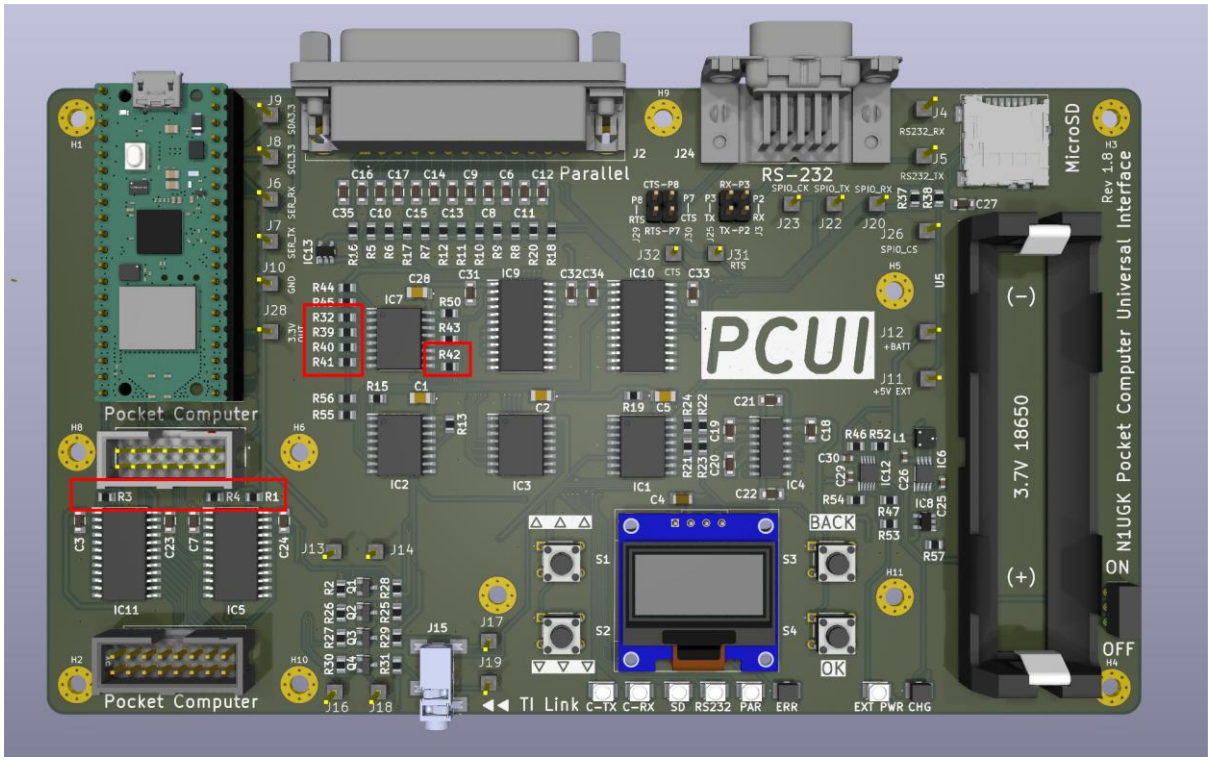


- Solder L1 (EPL3015-472MLC) then test the connection across the two pins of IC6 as shown below. It should indicate a "short", meaning the inductor is correctly soldered and making contact. **Avoid using too much heat on this inductor, and avoid touching the case with the iron, as it will melt.**

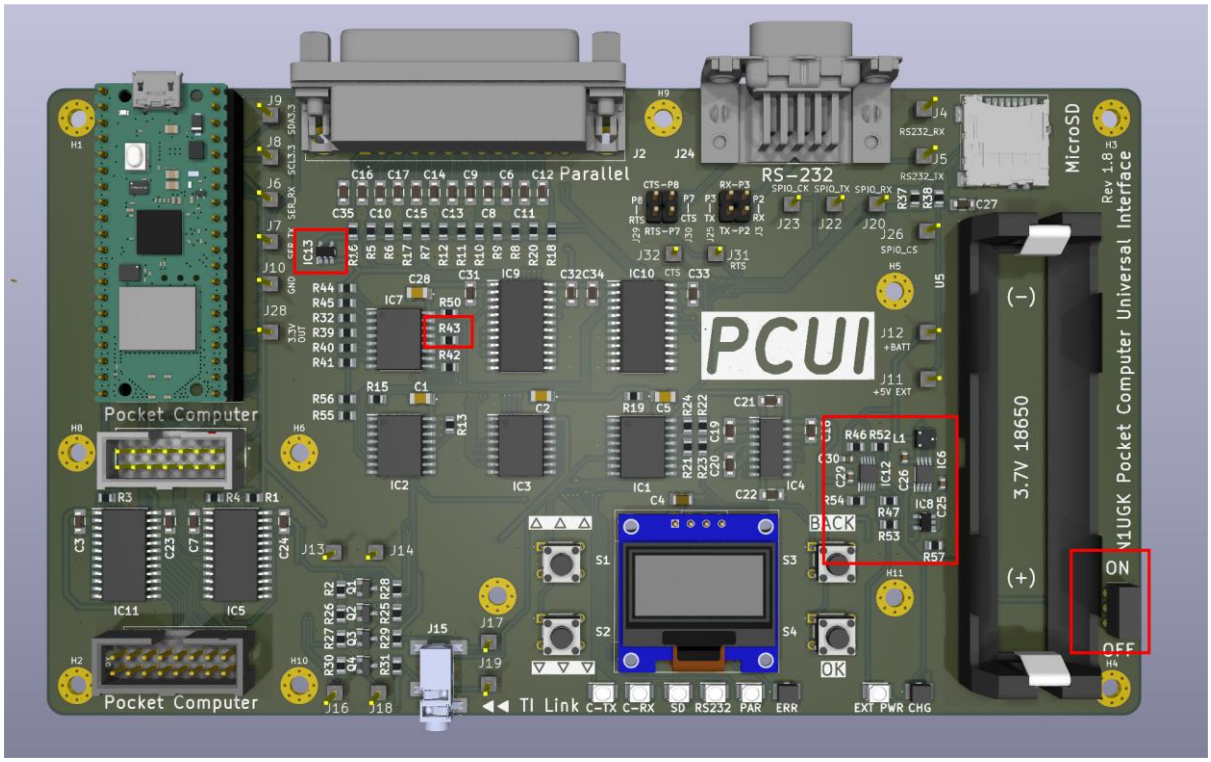
I have found it easiest to place solder paste on one pad on the PCB, then position the inductor so that only about ½ of the surface of each leg (underside) is on the pads, exposing a good portion of the pads on the PCB for the iron to heat up effectively, pressing lightly down on the inductor when heating the pad with the solder paste. Then, once solidly affixed on one side, heat the other pad, add solder, and the solder should travel across the pad to the leg of the inductor.



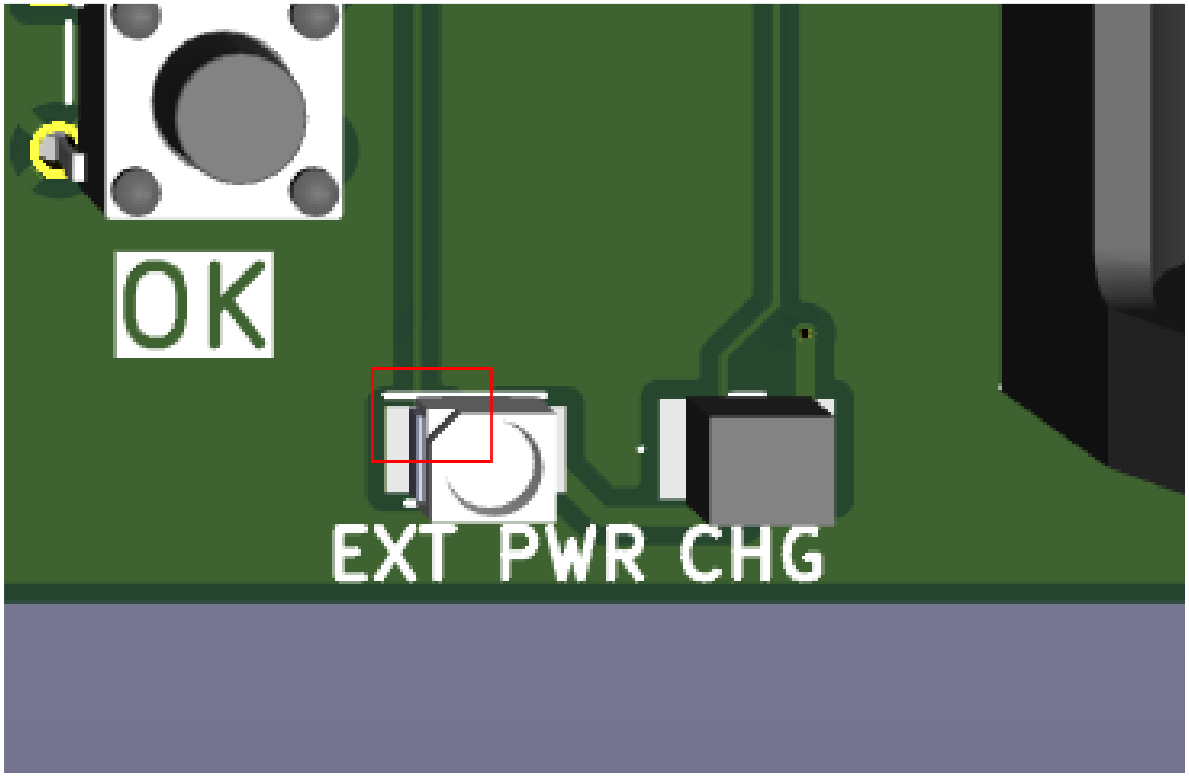
- Solder IC6 (BQ24090DGQR)
- Solder IC12 (MCP1642B-50I/MS)
- Solder IC8 and IC13 (MAX40200AUK+T Ideal Diode)
- Solder 2 150 ohm resistors R43, R52
- Solder 2 10K resistors R46, R57
- Solder 9 3.3K resistors R47, R01, R03, R04, R32, R39, R40, R41, R42, as shown in the image below. R47's location is shown in the image above.

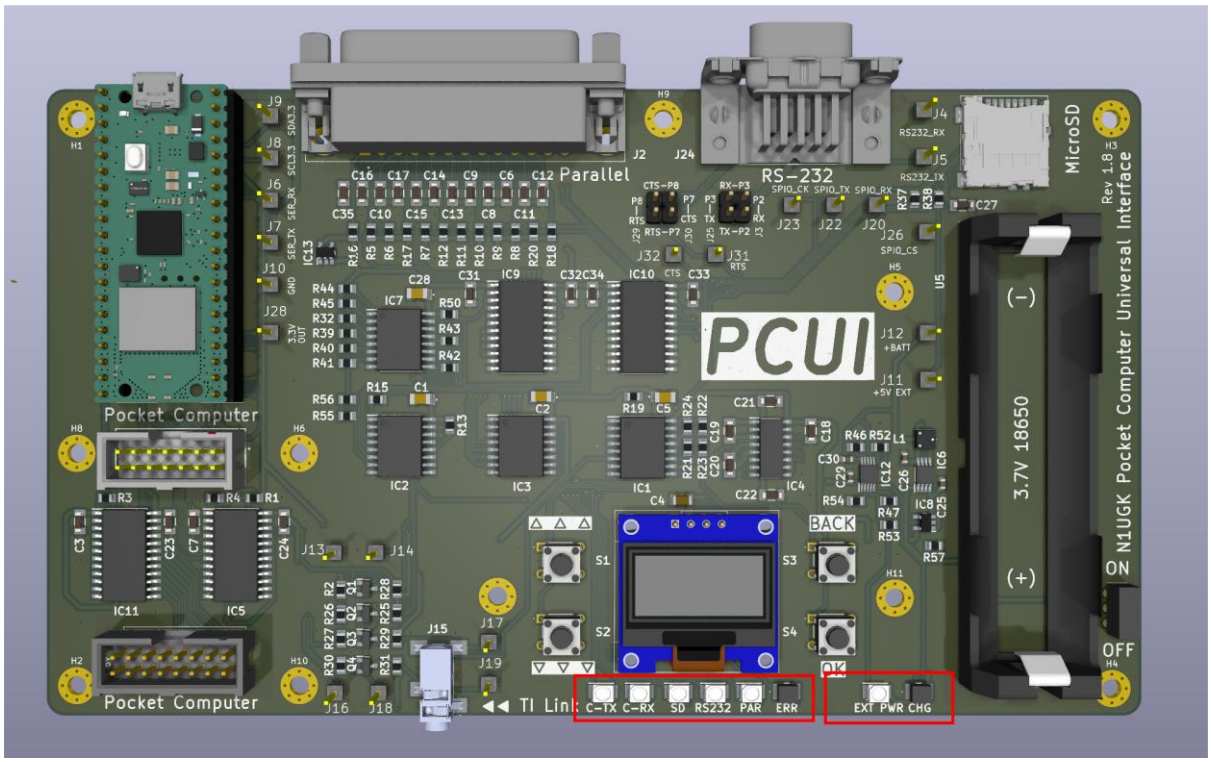


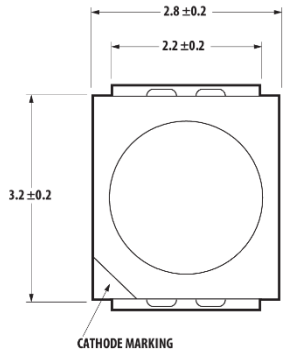
- Solder 1 2K resistor R53
- Solder 1 1K resistor R54
- Solder 4 10 uF capacitors C26, C25, C29, C30.
- Solder S5 (power switch).



- Solder 6 green LEDs: EXT PWR, C-TX, C-RX, SD, RS232, PAR pads
- Solder 2 red LEDs on the CHG and ERR pads, and note the orientation – the notch to the left as shown below:







Note the cathode side is marked with a notch on the corner as shown in the image below. The cathode is the negative or GND side of the LED. **This should point to the LEFT on the PCB for all LEDs**.

- Solder a header pin to J12, J11, and J10.

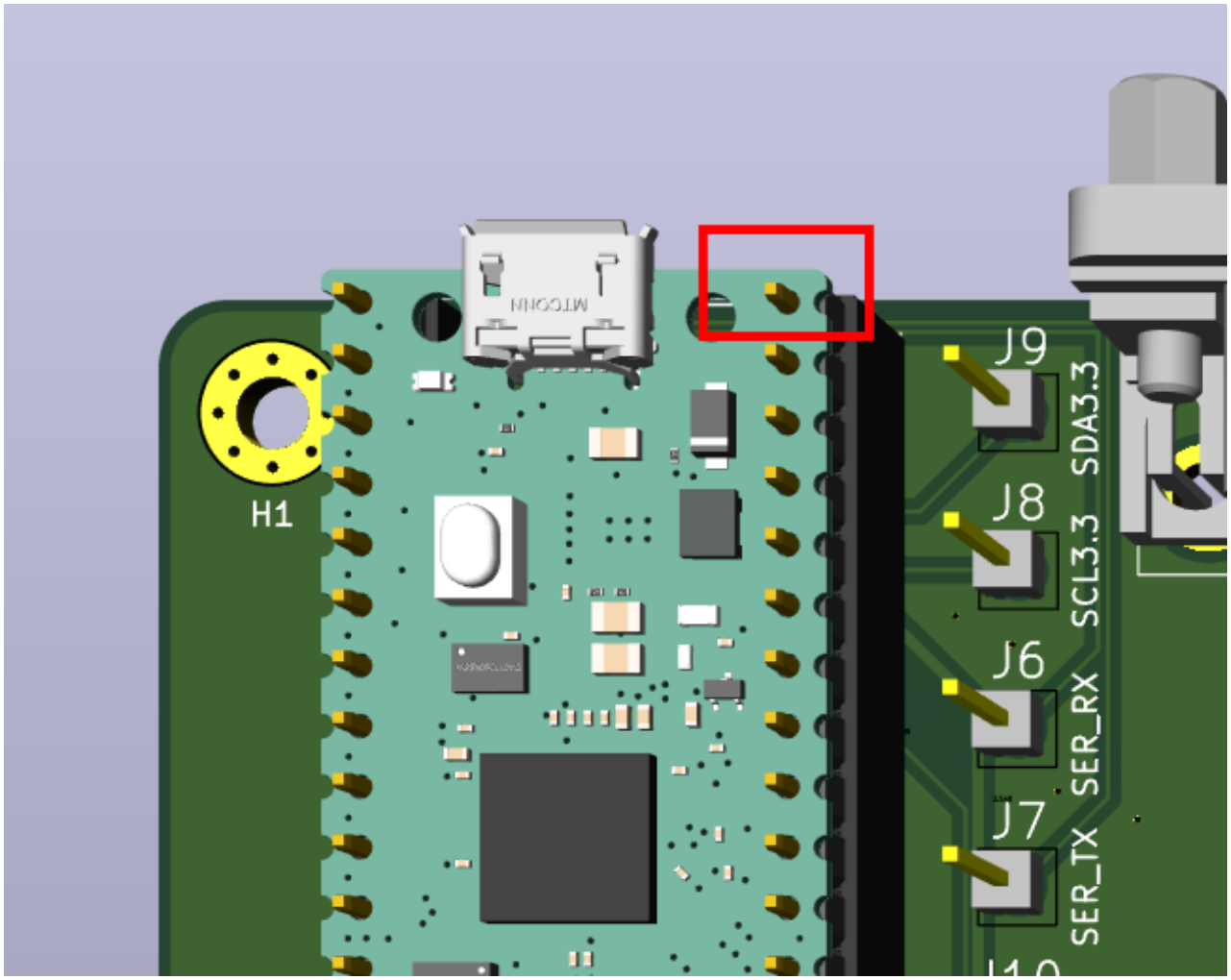
2. Testing

Apply 3.7 volts DC to pin J12, and limit current to about 100mA. Use J10 as GND. Turn the power switch S5 on the PCUI ON. Measure the voltage from pin J11. You should see 5 volts DC on pin J11.

If you do not see 5 volts and excessive current, look for a solder bridge on the ICs you soldered. If you do not see 5 volts and no current draw, double check L1 and the capacitors you soldered – if there's an "open" connection on any parts, the power supply will not output voltage. You can lightly press on each component and pin for example to check for missing connections or bad solder joints.

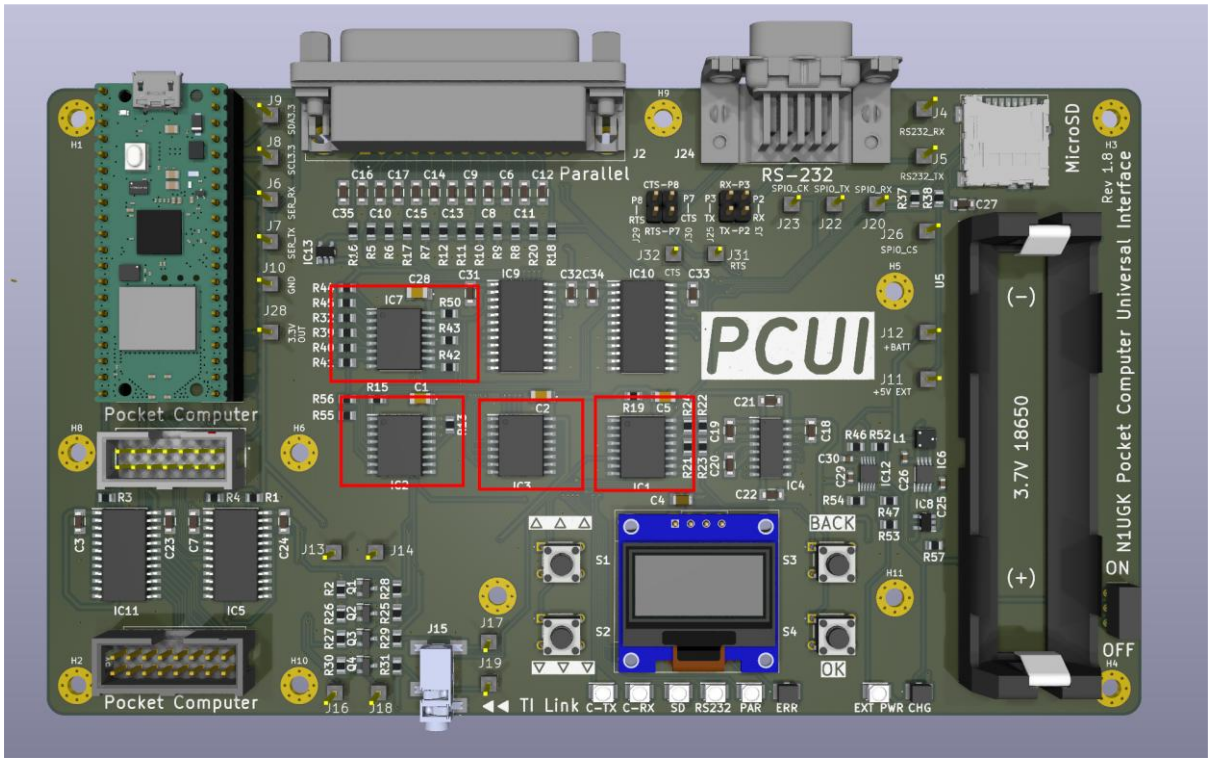
Once you get the DC/DC step up working (+5 volts from J11), connect your bench power supply to the top right most pin of the Pico board, which is VBUS. Set the supply to 5 volts DC, with a current limit of 100 mA. Use J10 for GND. Turn on the bench supply and check that the green EXT PWR light illuminates.

If the EXT PWR light does not illuminate, make sure the orientation of the LED is correct, and then check IC12 for solder bridges or pins without a good connection. Also check capacitors and resistors to make sure they are well connected.


H1
J9
J8
J6
J7
SDA3.3
SCL3.3
SER_RX
SER_TX

B. MAX 7329 AWE Port Expanders

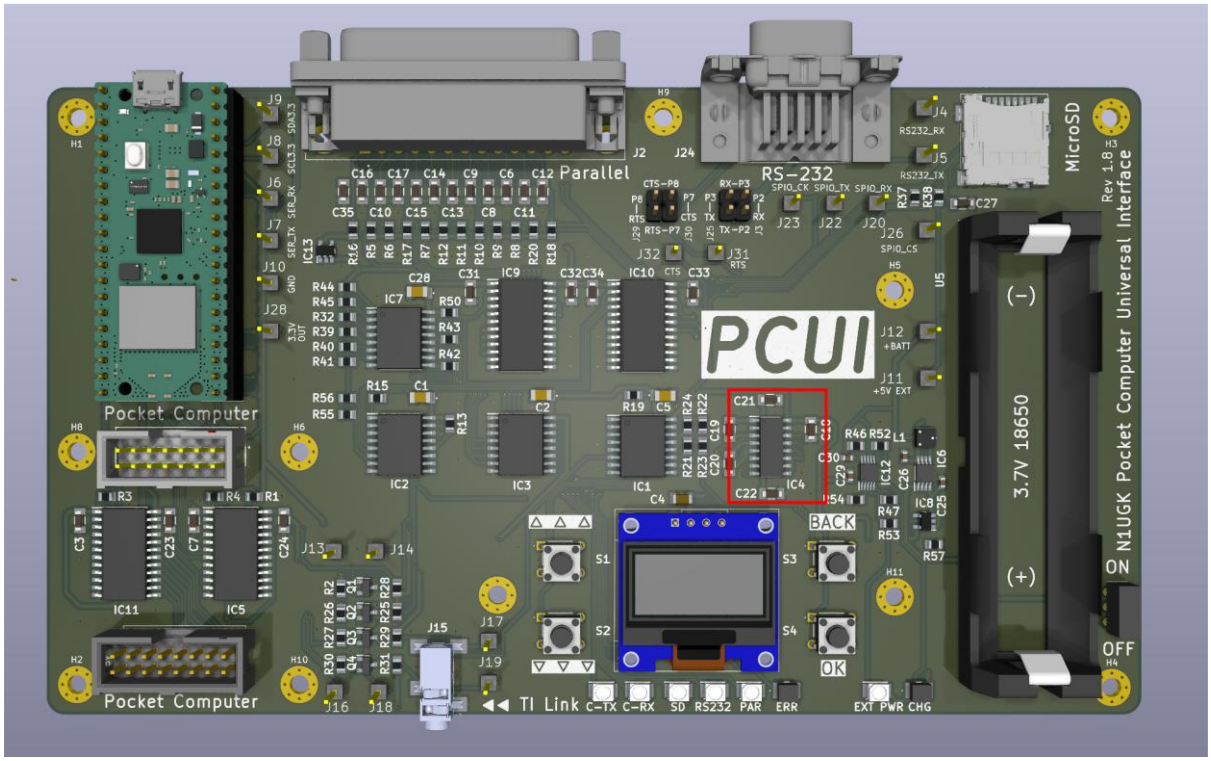
- Solder 4 MAX 7329 AWE Port Expanders IC1, IC2, IC3, IC7:



C. SP3232 RS-232 Interface IC

- An SP3232 RS-232 interface IC needs to be soldered onto the PCB where the silkscreen indicates IC4 (see highlighted area below):

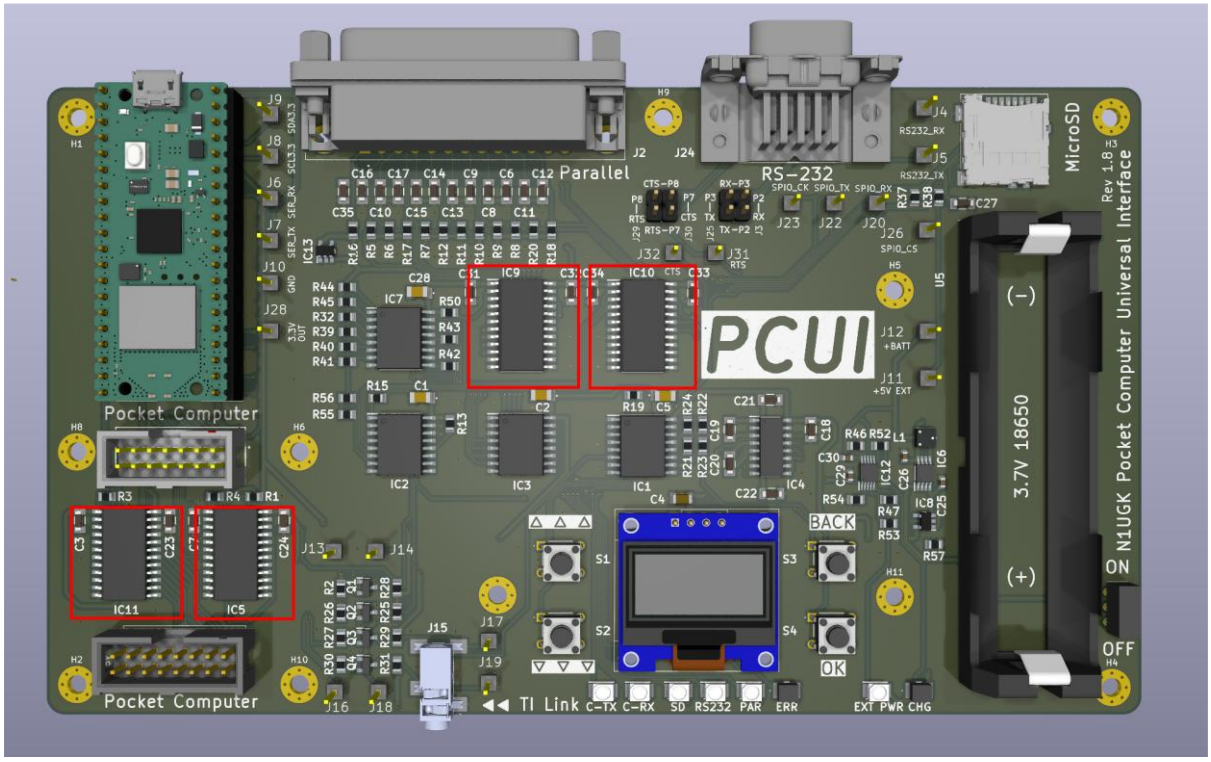
Be sure to note the orientation of the SP3232 IC – Pin 1 is indicated on the silkscreen on the top right corner with a horizontal line.



D. SN74LVC8T245DWRG4 Buffers

- 4 SN74LVC8T245DWRG4 buffer ICs need to be soldered onto the PCB where the silkscreen indicates IC11, IC5, IC9, and IC10 (see highlighted area below):

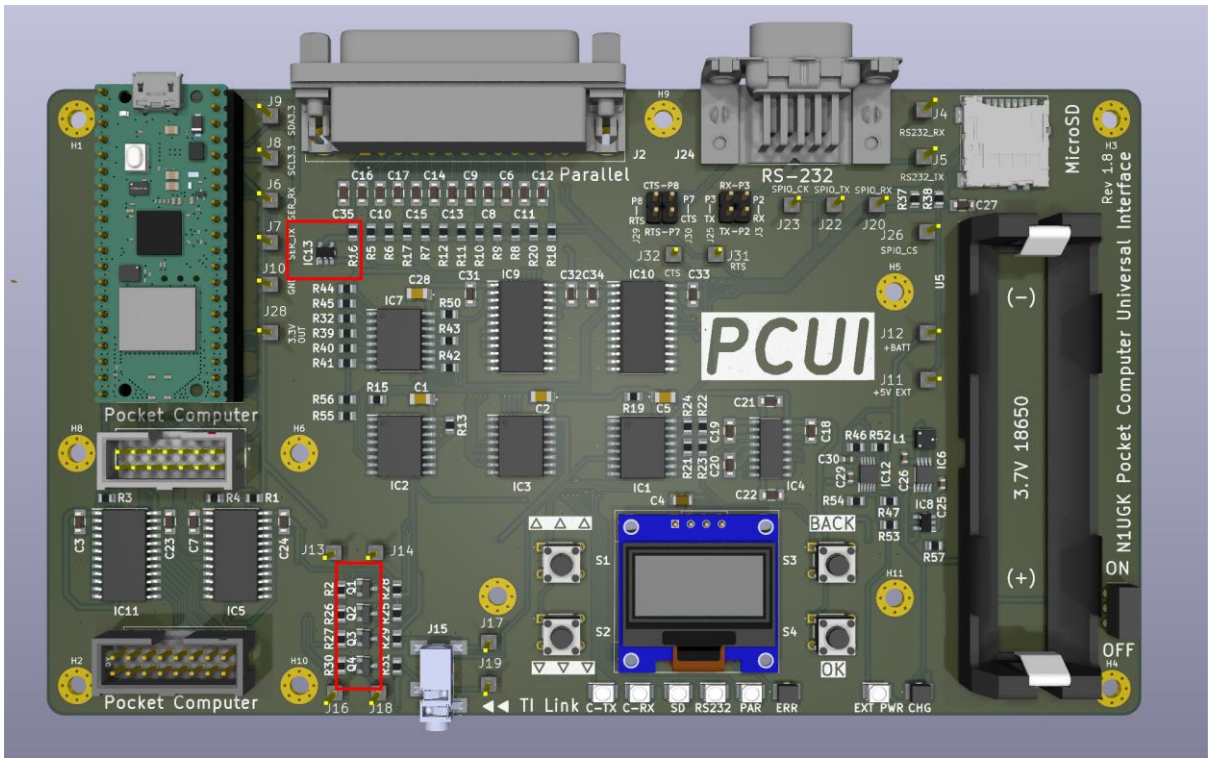
Be sure to note the orientation of the SN74LVC8T245DWRG4 IC – Pin 1 is indicated on the silkscreen on the top right corner with a horizontal line.



E. BSS138s

- Solder 4 BSS138s for Q1, Q2, Q3, and Q4.

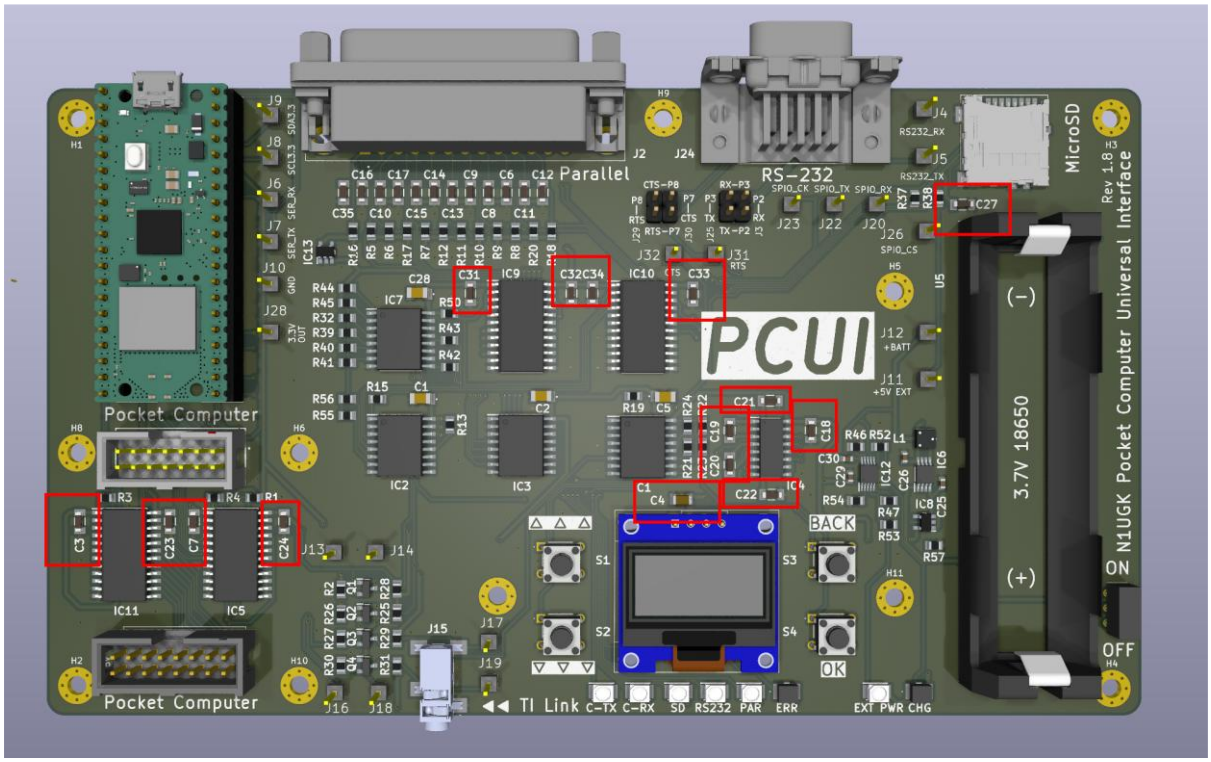
See highlighted area below:



F. 0.1 uF SMD Capacitors

- Solder 15 0.1 uF capacitors C03, C04, C07, C18, C19, C20, C21, C22, C23, C24, C27, C31, C32, C33, C34

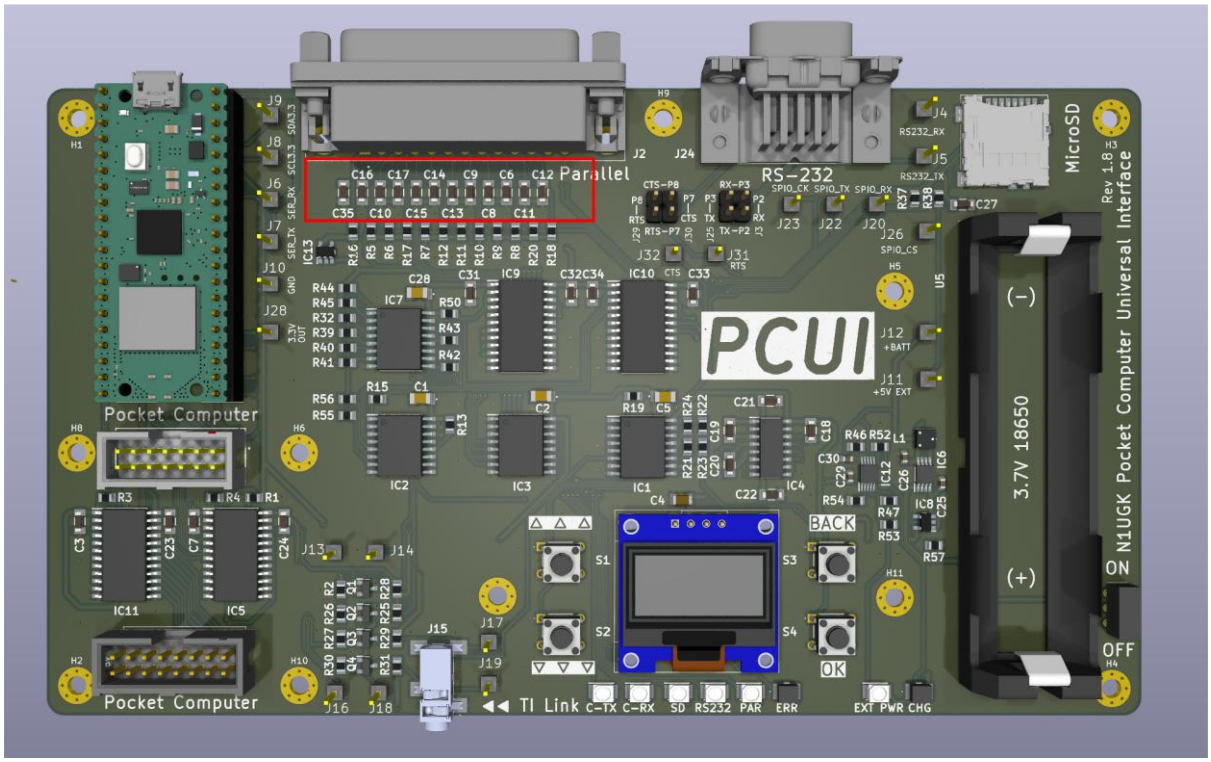
See highlighted areas below:



G. 470 pF SMD Capacitors

- Solder 12 470 pF capacitors C06, C08, C09, C10, C11, C12, C13, C14, C15, C16, C17.

See highlighted areas below:

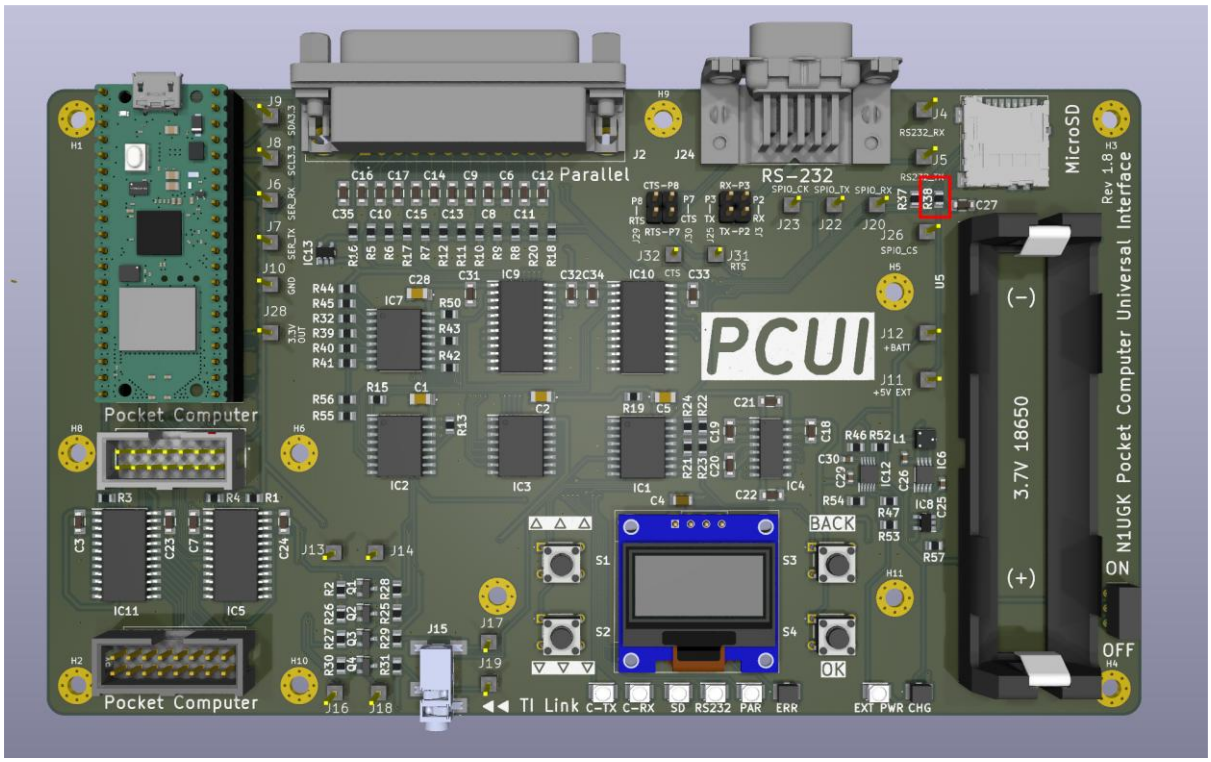


V. PCUI Main Board Assembly Instructions

A. 100 K SMD Resistor

- Solder 1 100K resistor R38.

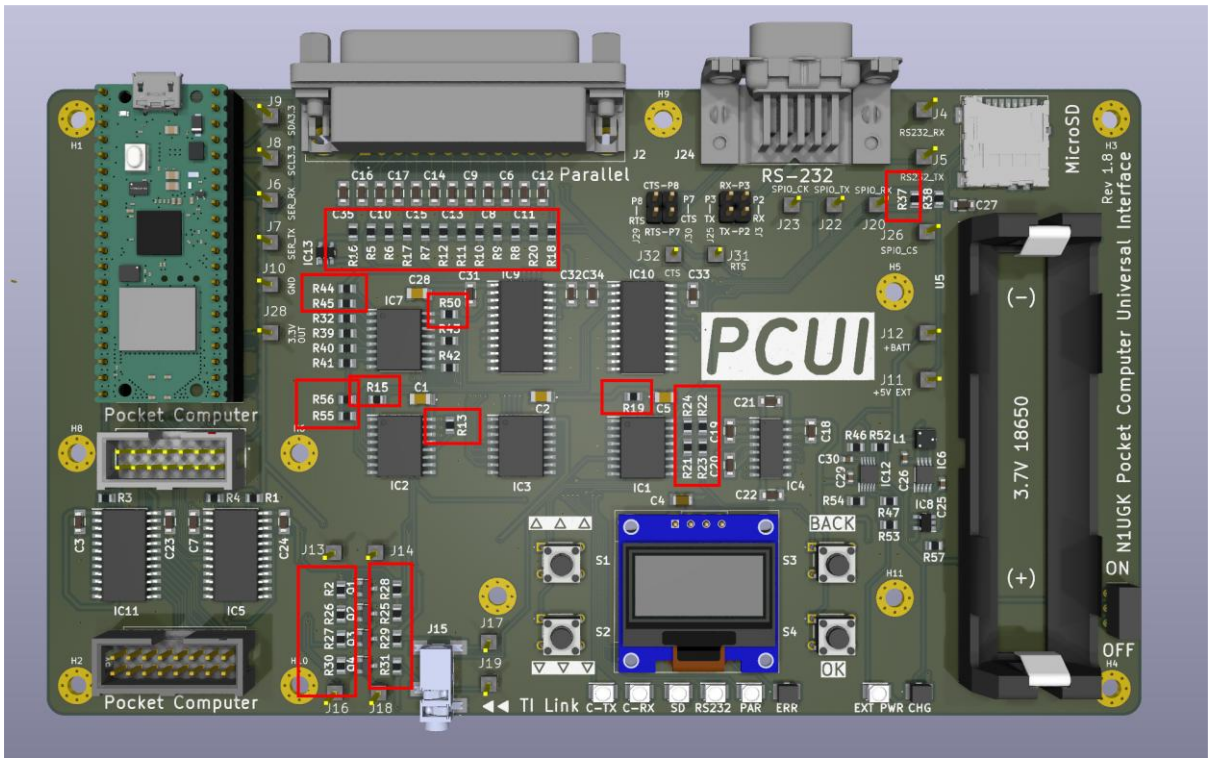
See highlighted areas below:



B. 10 K SMD Resistors

- Solder 33 10K resistors R02, R05, R06, R07, R08, R09, R10, R11, R12, R13, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R37, R44, R45, R50, R55, R56.

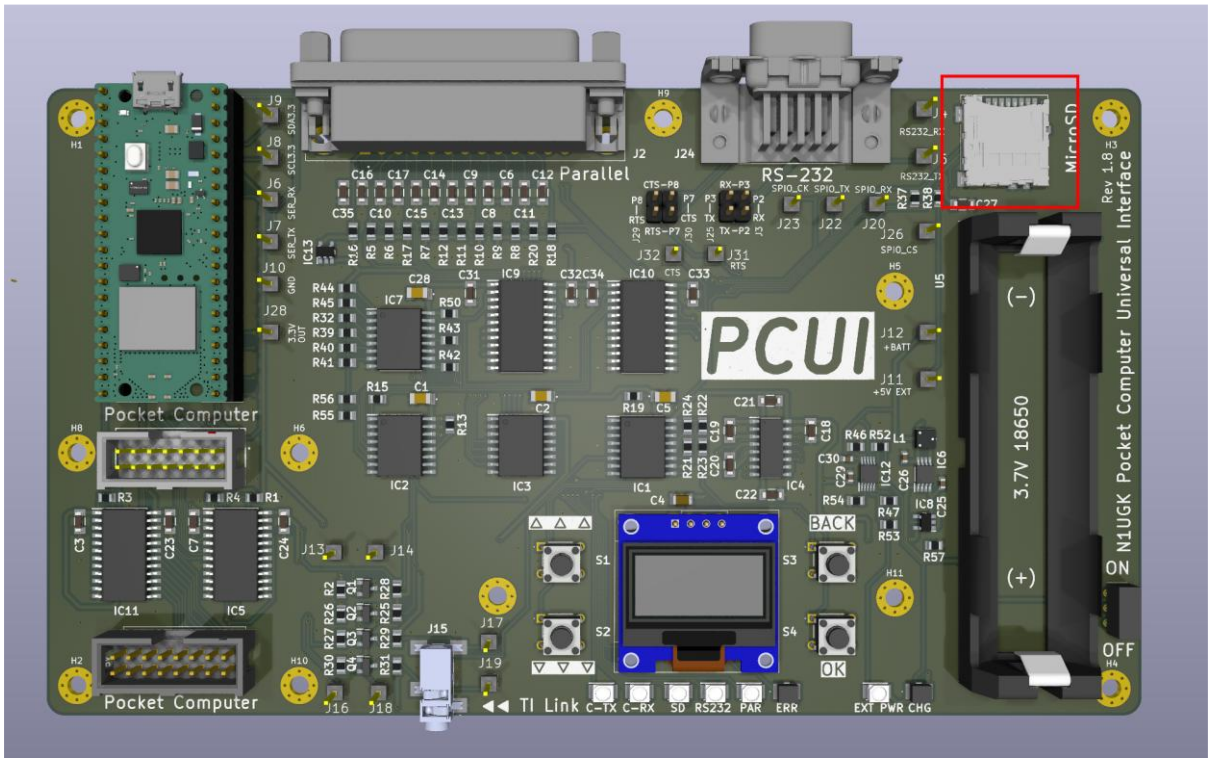
At this point in the assembly, all other resistor footprints/pads remaining are 10K. See highlighted areas below:

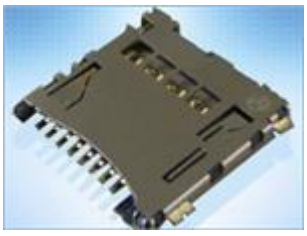


C. Micro SD

- Solder the micro SD holder.
- Insert the micro SD card into the slot after it is soldered.

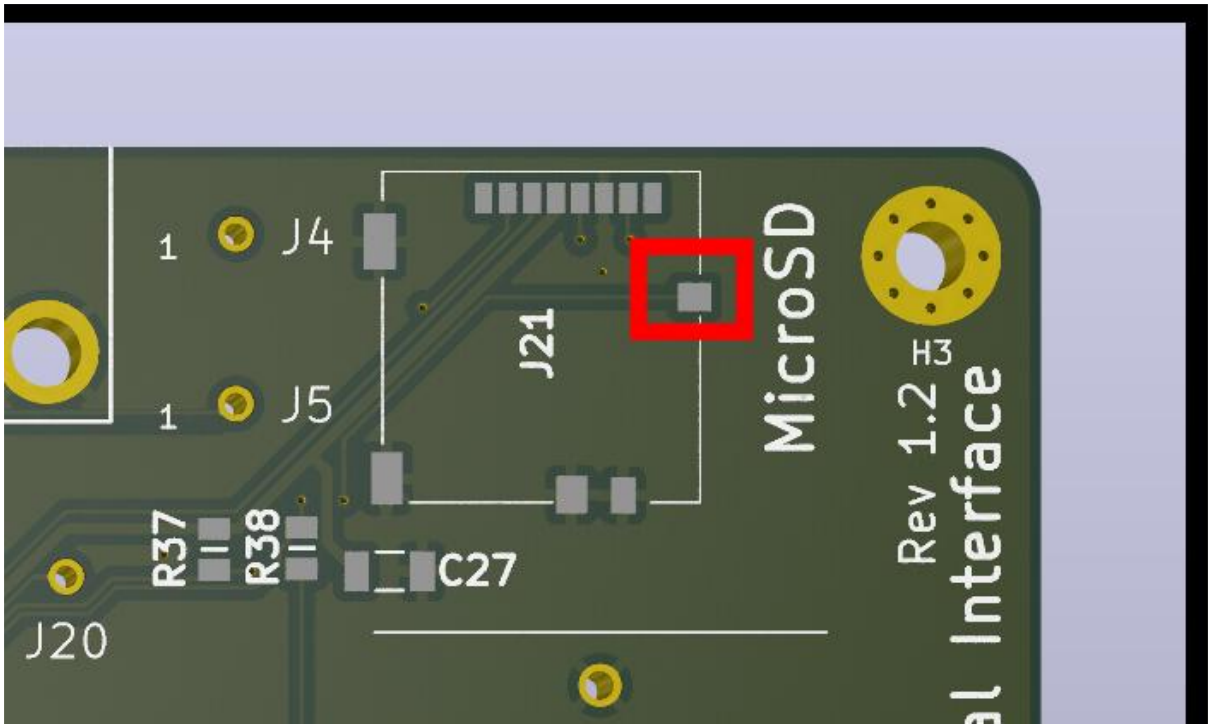
See highlighted area below:



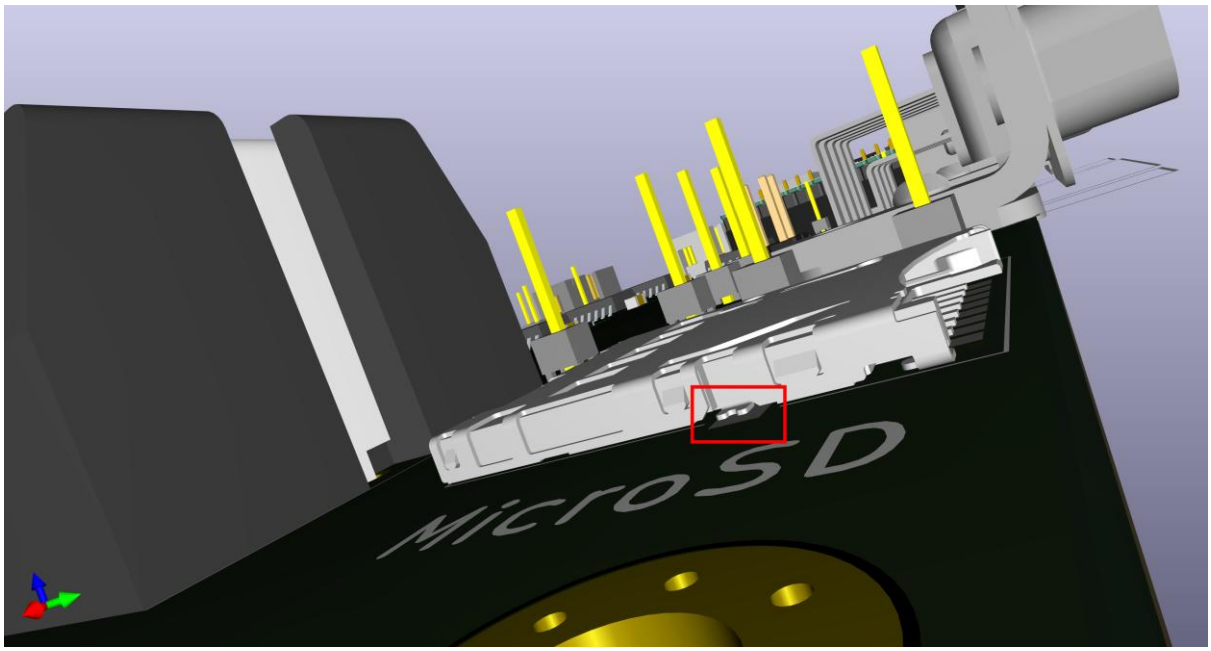


A micro SD holder needs to be soldered onto the PCB in the area labelled J21. **Be extremely careful in soldering this holder onto the PCB**. Use very little solder, as little as possible on the pin connections. Too much solder will result in clearance issues with the SD card. Solder paste would be recommended.

Also note the card switch contact highlighted below. Use very little solder here, as this pin connector is very close to the metal housing which is grounded. If too much solder is used here, you will end up with a solder bridge to the housing and grounding this signal line. If this happens, SD card detection will not work properly, and the SD card functions will be disabled.



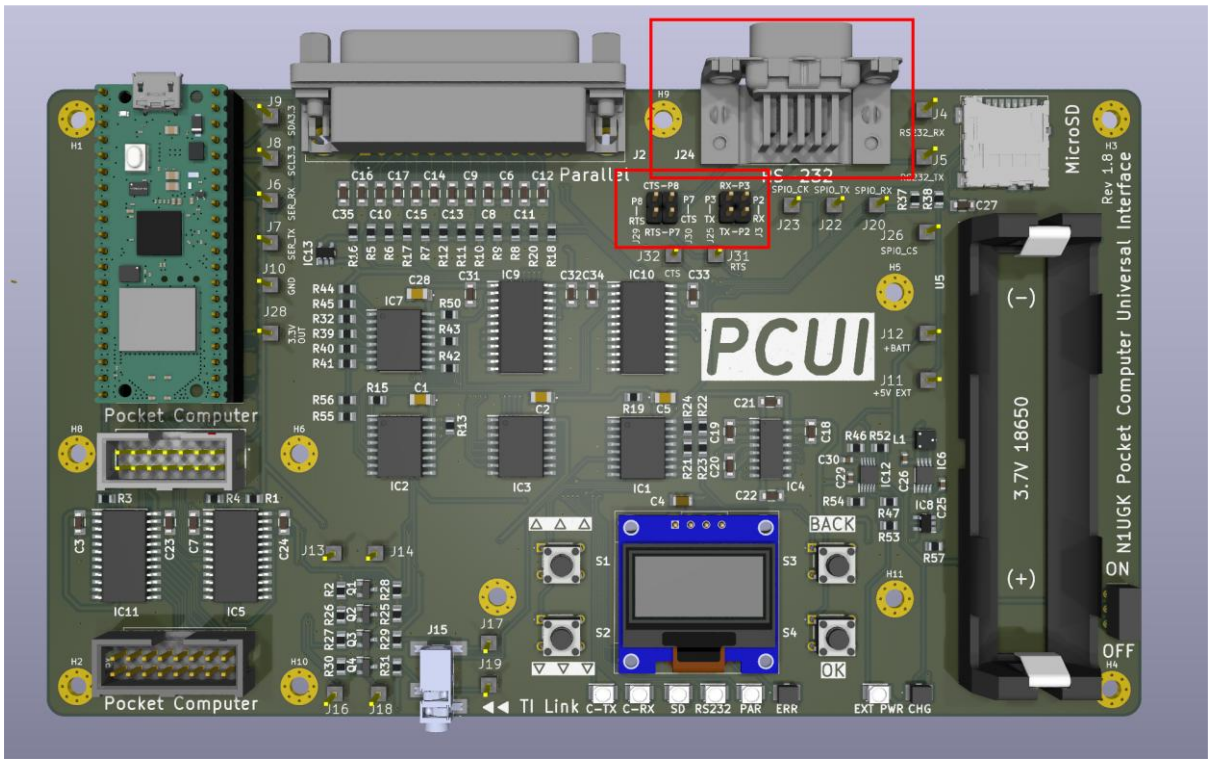
The switch pin is shown below. **Too much solder here will bridge the pin with GND, effectively disabling the SD card functions**, as the PCUI would detect no card inserted.

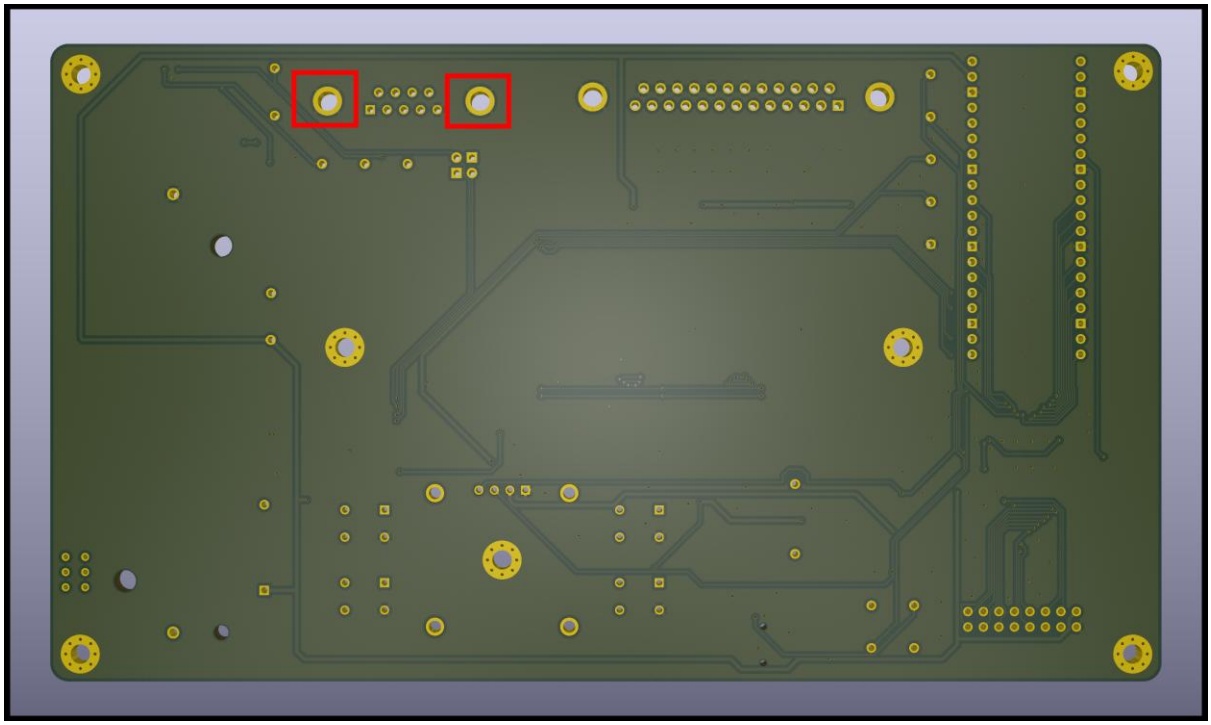


D. RS-232 Jumpers and RS-232 DB-9 Connector

- Solder jumpers for RTS/CTS and RX/TX for the RS-232 port.
- Place jumpers in a vertical position over the RX/TX section, and leave the jumpers off for the RTS/CTS flow control.
- Solder the DB-9 connector onto the PCB. Be sure to solder all pins in order to provide adequate grounding / shielding as well as mechanical stability. Be sure to solder the metal binding posts as well. Soldering the binding posts will provide mechanical stability and prevent damage to the PCB and the connector from forces applied when plugging and unplugging connectors to and from this port.

See highlighted areas below:

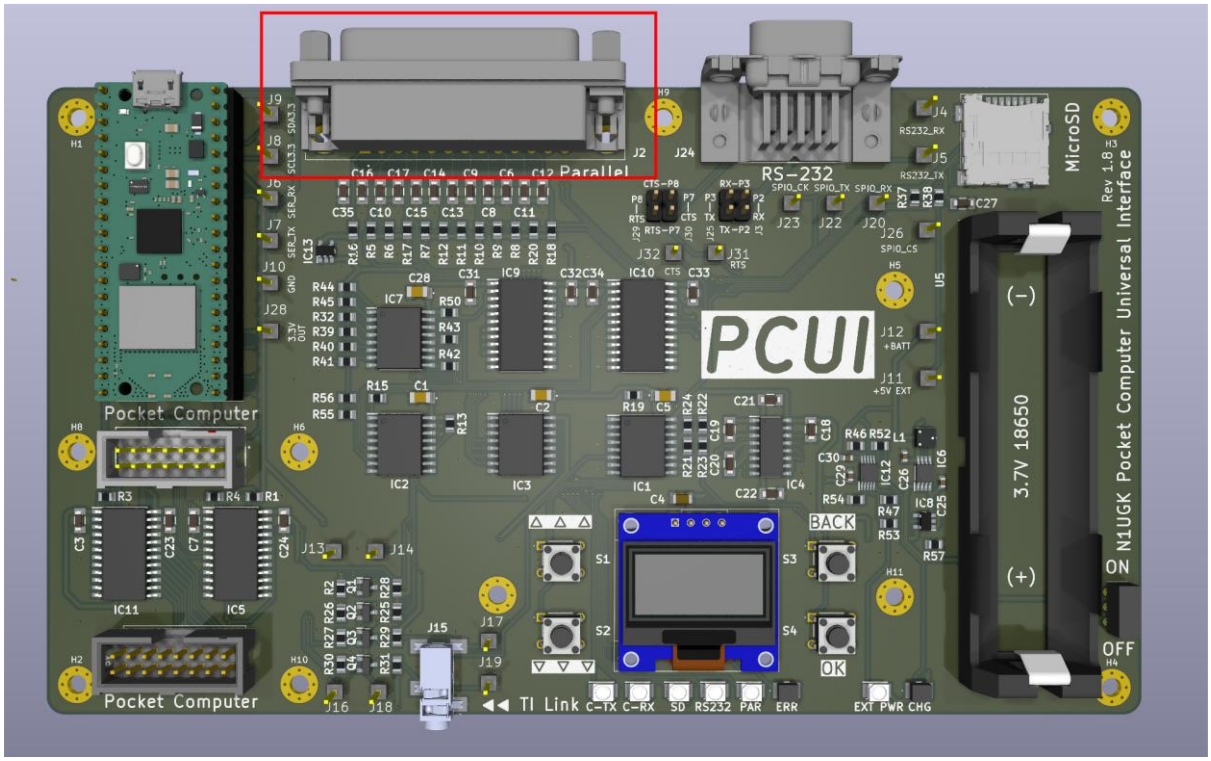


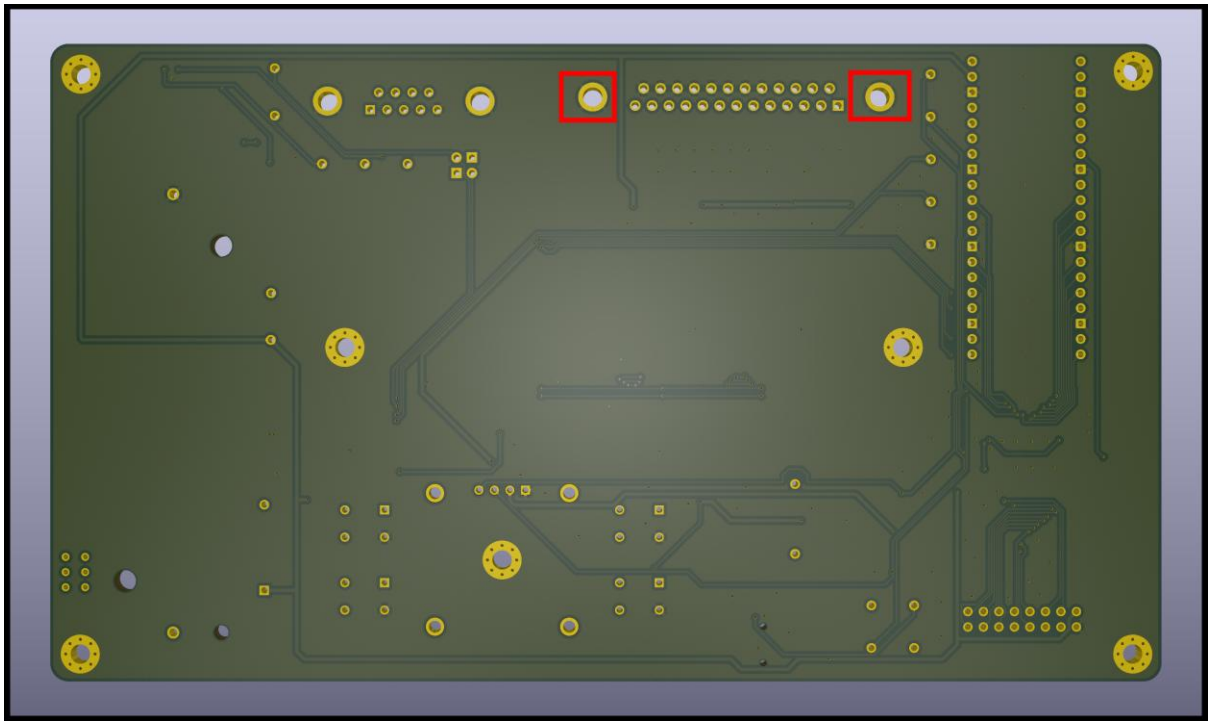


E. Parallel Port Connector

- Solder the 25-pin connector onto the PCB. Be sure to solder all pins in order to provide adequate grounding / shielding as well as mechanical stability. Be sure to solder the metal binding posts as well. Soldering the binding posts will provide mechanical stability and prevent damage to the PCB and the connector from forces applied when plugging and unplugging connectors to and from this port.

See highlighted areas below:



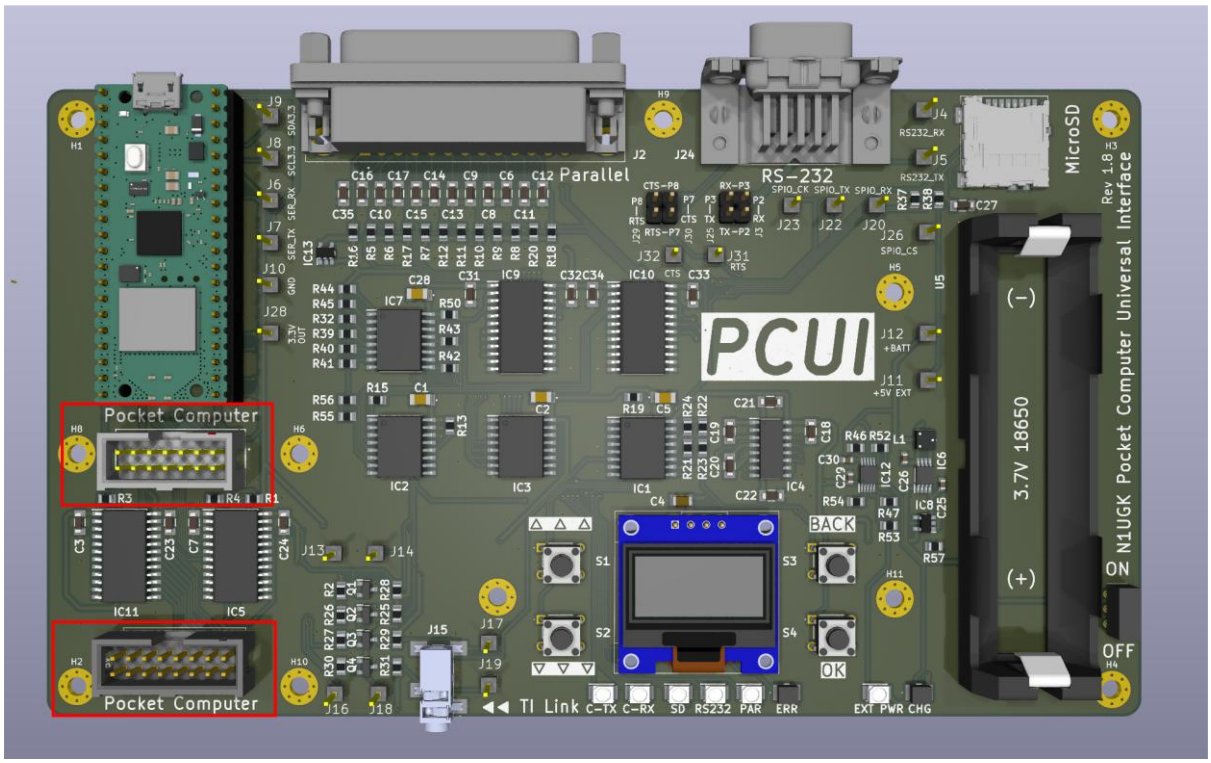


F. Pocket Computer 14 and 16-pin Connectors

- Solder the 14-pin IDC header for J27
- Solder the 16-pin IDC header for J1.

Be sure the "notch" faces upward toward the top of the PCB.

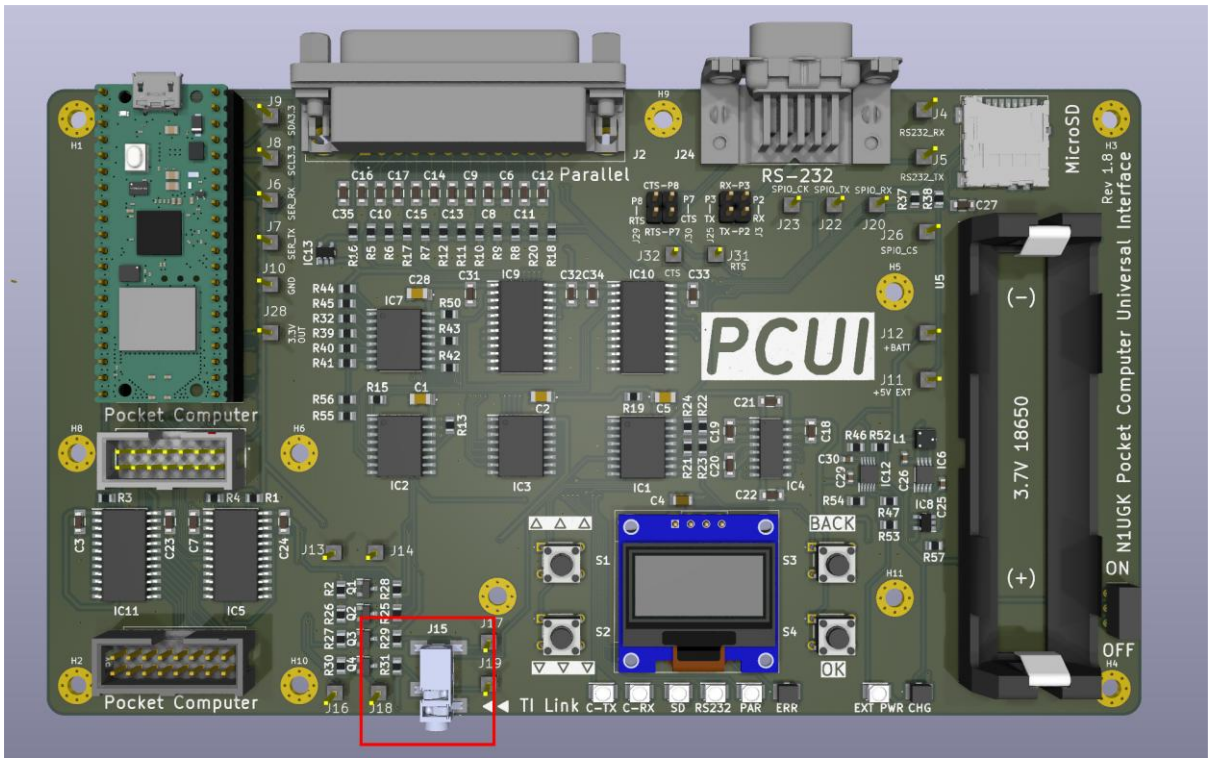
See highlighted areas below:



G. TI-Link Connector

- Solder the 3.5mm stereo connector for J15.

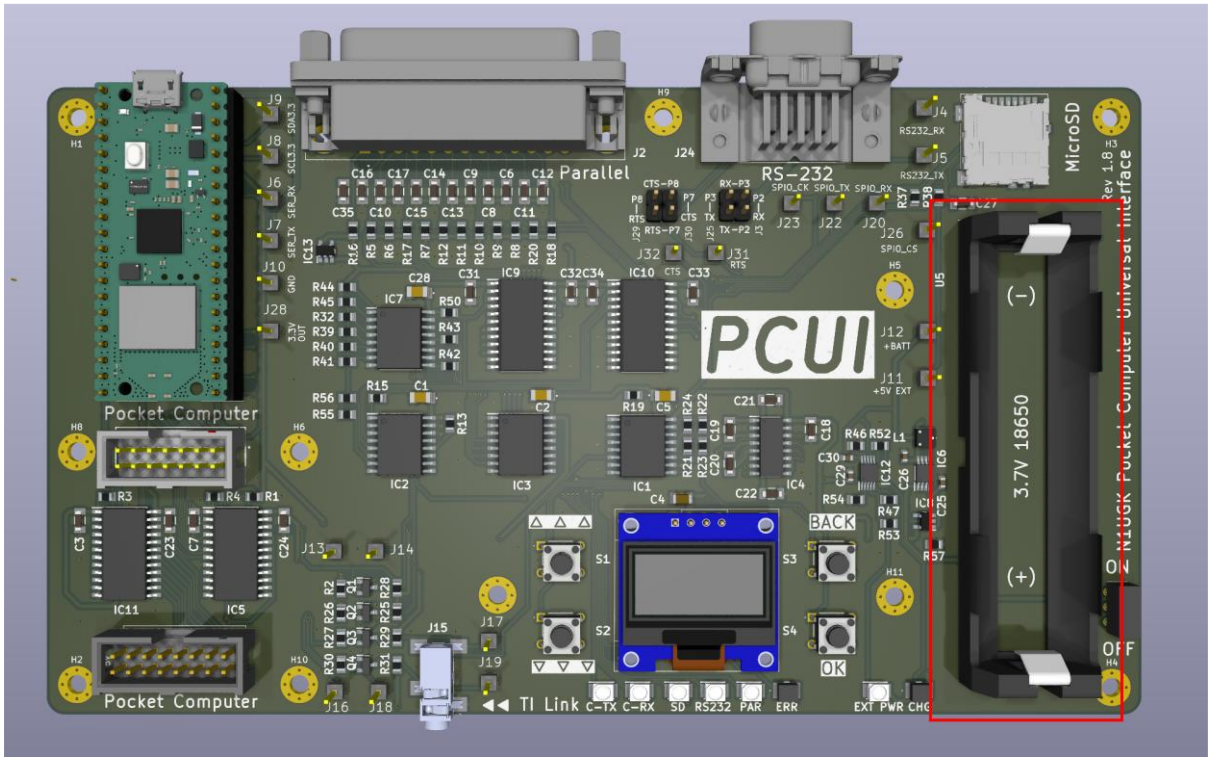
See highlighted area below:



H. 18650 Battery Holder

- Solder the 18650 battery holder onto the PCB. Note, it may be necessary to sand off the surface a bit on the metal pins that you will be soldering to the PCB – if solder does not adhere. *Do not add an 18650 battery at this point.*

See highlighted area below:

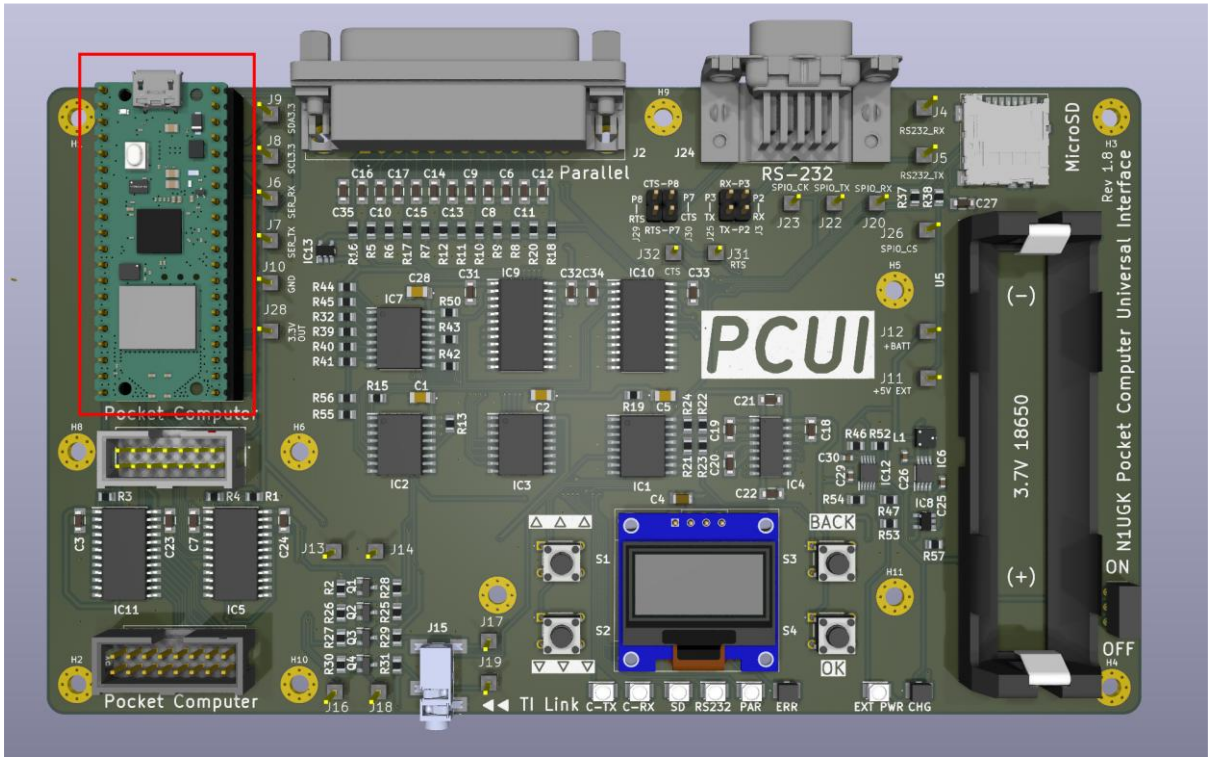


I. Pico 2 W

- Solder the Pico 2 W onto the PCB. Note that the leads should be trimmed so that they do not protrude more than 2-3mm below the PCB. This ensures the standoffs will sufficiently keep the pins away from any surfaces.

Be sure the USB connector on the Pico 2 faces the top of the board.

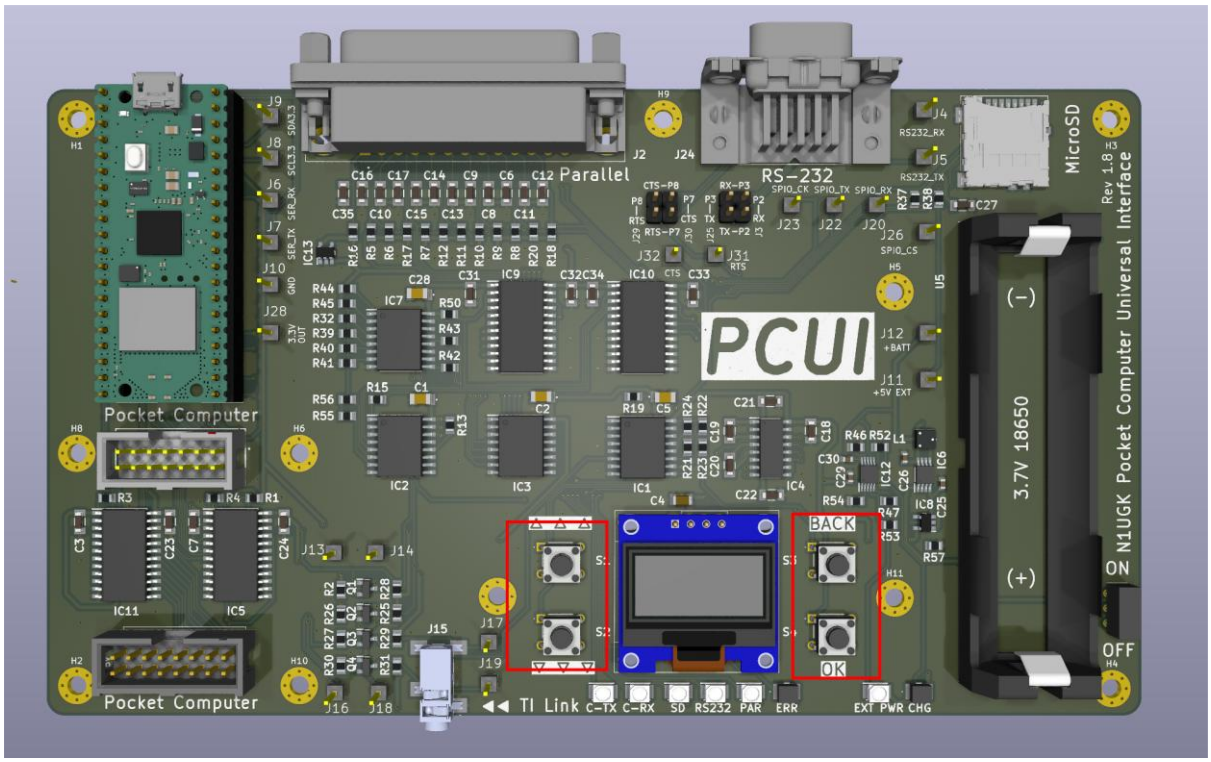
See highlighted area below:

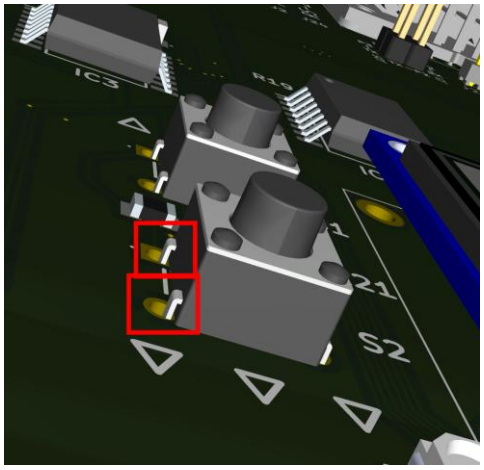


J. Micro Push Buttons and OLED

- Solder 4 micro pushbutton switches S1, S2, S3, S4.

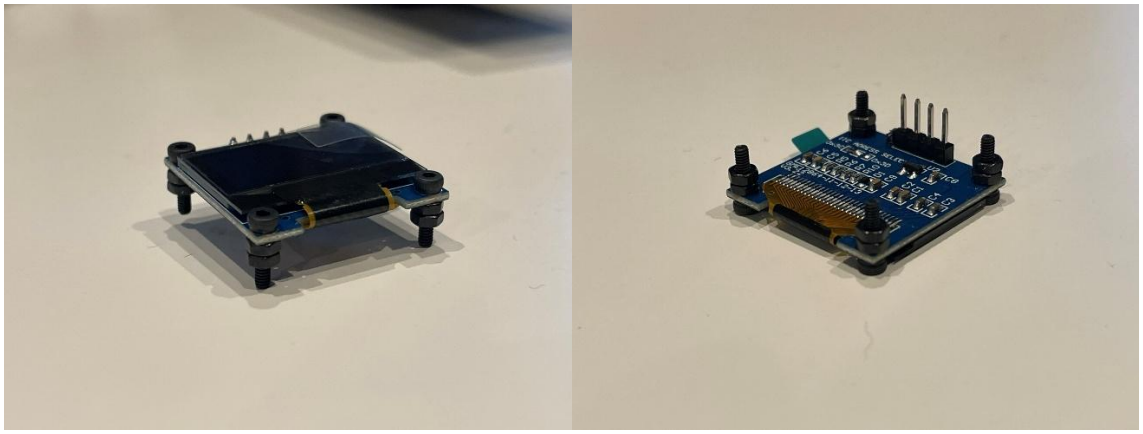
See highlighted areas below:





Note that the pins on these micro switches should be oriented so that they are on the left and right sides of the switch (not the top and bottom of the switch)

Mounting bolts for the OLED need to be attached – use two nuts on each bolt (underside). For example, see pictures below:



These nuts add spacing needed for leveling of the OLED against the PCB.

Then fasten the OLED onto the PCB with the 4 remaining nuts, then solder the 4 pins.

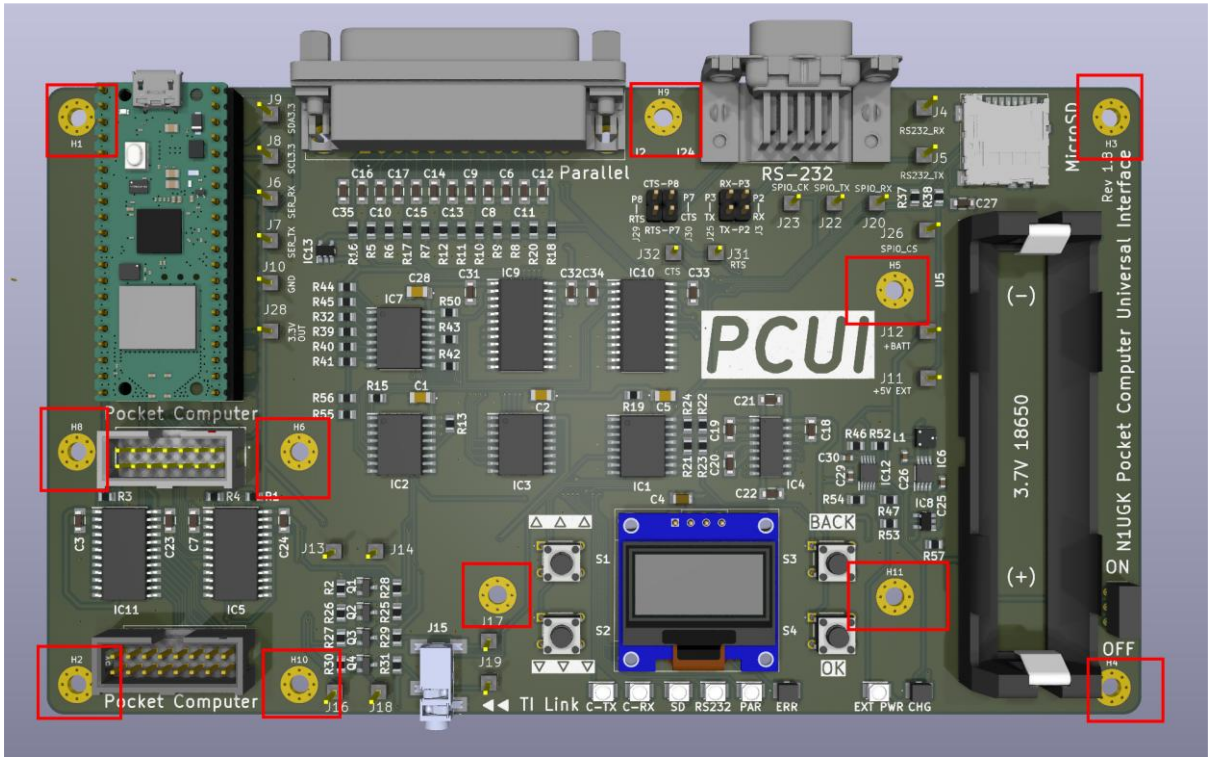
K. Nylon Standoffs



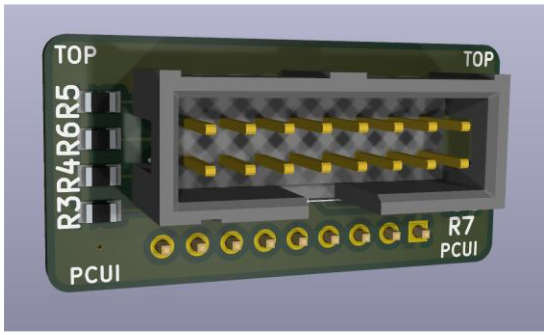
If you ordered a case for the PCUI, skip this step.

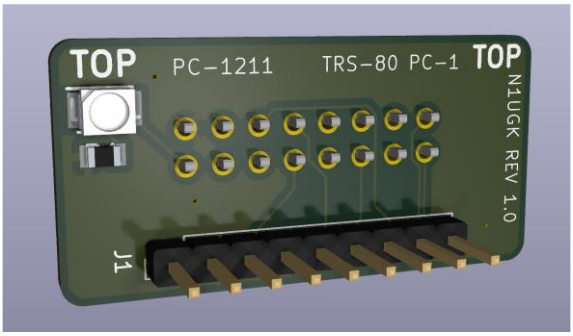
Install 11 nylon standoffs for H1, H2, H3, H4, H5, H6, H7, H8, H9, H10, H11.

See highlighted areas below:



VI. TRS-80 PC-1 / Sharp PC-1211 Interface Pack





The TRS-80 PC-1 / Sharp PC-1211 interface pack is a double sided PCB, meaning there are components on both sides.

A. 10K SMD Resistors

- Solder 5 10K SMD resistors R01, R03, R04, R05, R06, R07.

B. Green SMD LED

- Solder a green SMD LED, LED01. Note the notch on the LED should point to the left.

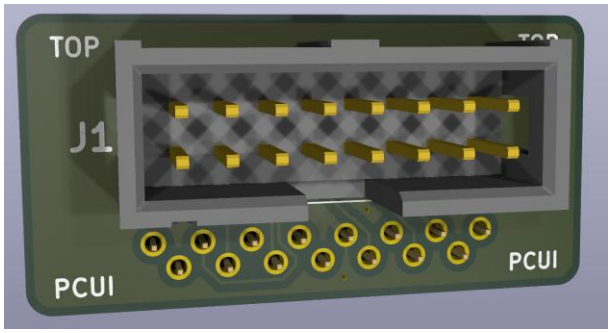
C. 16-Pin IDC Connector

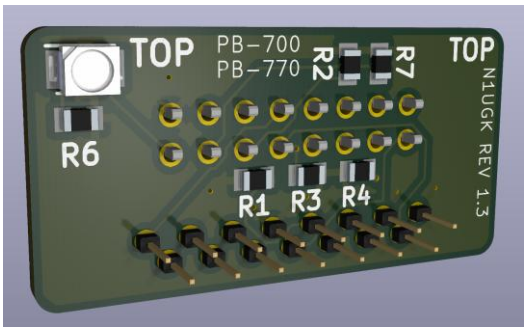
- Solder a 16-pin IDC connector J2. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side with the 4 resistors, not the side with the 1 resistor and 1 LED. Note the notch orientation points downward.

D. 9-Pin Pocket Computer Connector

- Solder the 9-pin pocket computer connector J1. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side with the 1 LED and 1 resistor, not the side with 4 resistors and the J2 connector.

VII. Casio PB-700/PB-770 Interface Pack

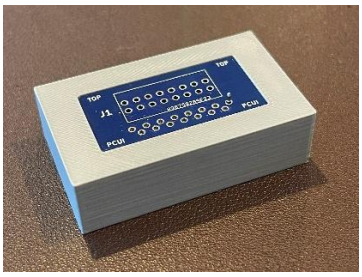




The Casio PB-700 and PB-770 interface pack is a double sided PCB, meaning there are components on both sides.

A. 15-Pin Pocket Computer Connector

- Solder the 15-pin pocket computer connector. Be sure to solder this connector on the correct side of the PCB. It is to be soldered so that the pins come out on the side with the 1 LED and resistors, meaning the soldering happens on the same side as connector J1.
- Use the 3D printed mold to place the PCB with J1 upward. Then place the 15 pins straight down so that they protrude just a few millimeters above the surface of the PCB – then solder in place. **Note: Remove the plastic base on these pins using a nibbing tool or fine wire cutter tool.**



B. 10K SMD Resistors

- Solder 6 10K SMD resistors R01, R02, R03, R04, R06, R07.

C. Green SMD LED

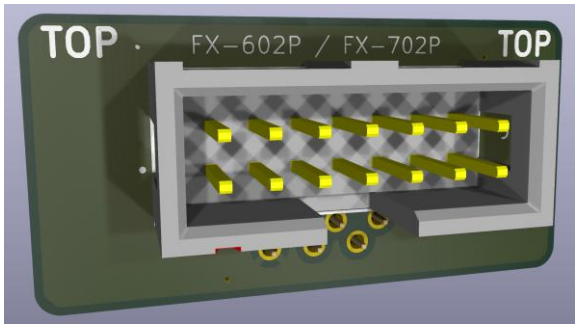
- Solder a green SMD LED, LED01. Note the notch on the LED should point to the left.

D. 16-Pin IDC Connector

- Solder a 16-pin IDC connector J1. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side without the resistors and 1 LED. Note the notch orientation points downward.

VIII. Casio FX-702P/FX-602P Interface Pack





The Casio FX-702P and FX-602P interface pack is a double sided PCB, meaning there are components on both sides.

A. 7-Pin Pocket Computer Connector

- Solder the 7-pin pocket computer connector. Be sure to solder this connector on the correct side of the PCB. It is to be soldered so that the pins come out on the side with the 1 LED and resistor, meaning the soldering happens on the same side as connector J27.
- Use the 3D printed mold to place the PCB with J27 upward. Then place the 7 pins straight down so that they protrude just a few millimeters above the surface of the PCB – then solder in place. **Note: Remove the plastic base on these pins using a nibbing tool or fine wire cutter tool.**
- 

B. 10K SMD Resistor

- Solder 1 10K SMD resistor R06.

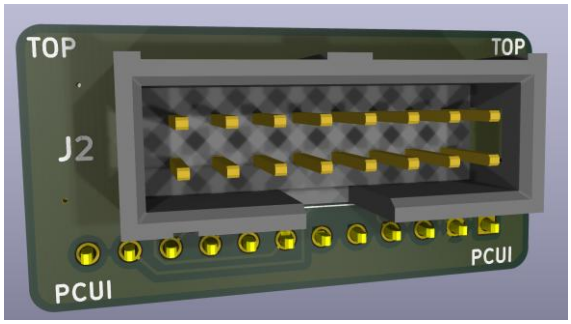
C. Green SMD LED

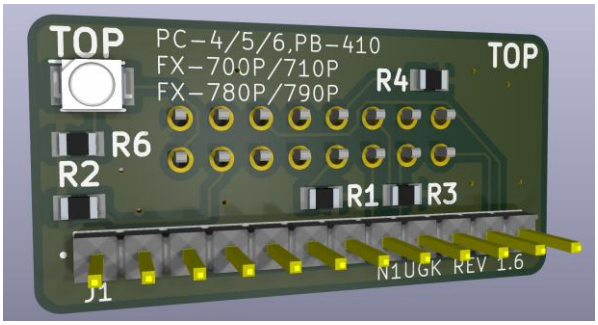
- Solder a green SMD LED, LED01. Note the notch on the LED should point to the left.

D. 14-Pin IDC Connector

- Solder a 14-pin IDC connector J27. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side without the resistors and 1 LED. Note the notch orientation points downward.

IX. Casio PB-100 Interface Pack





The Casio PB-100 interface pack is a double sided PCB, meaning there are components on both sides.

A. 10K SMD Resistors

- Solder 5 10K SMD resistor R01, R02, R03, R04, R06.

B. Green SMD LED

- Solder a green SMD LED, LED01. Note the notch on the LED should point to the left.

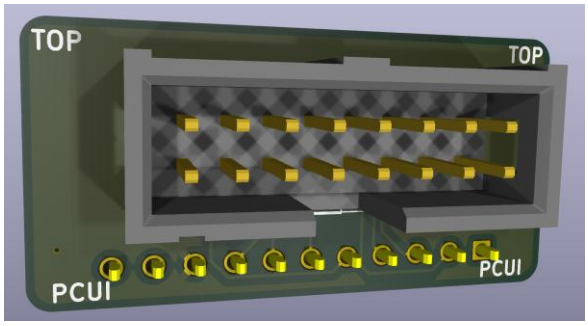
C. 12-Pin Pocket Computer Connector

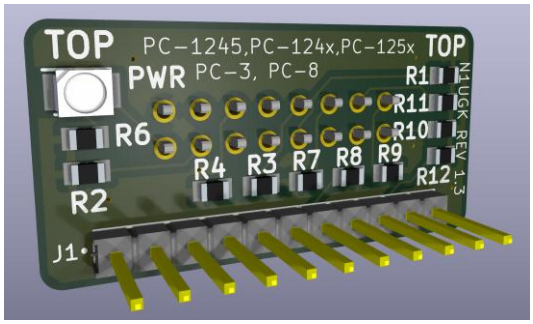
- Solder the 12-pin pocket computer connector. Be sure to solder this connector on the correct side of the PCB. It is to be soldered so that the pins come out on the side with the 1 LED and resistors, meaning the soldering happens on the same side as connector J2. *It is recommended to solder just 1 pin first, then ensure the connector pins are "level", adjusting the one soldered pin as needed. Once the connector is level, then solder the rest of the pins.*

D. 16-Pin IDC Connector

- Solder a 16-pin IDC connector J2. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side without the resistors and 1 LED. **Note the notch orientation points downward**, regardless of the silkscreen notation. *Rev 1.5 of the PCB incorrectly indicated the notch points upward.*

X. Sharp PC-1245 Interface Pack





The Casio Sharp PC-1245 interface pack is a double sided PCB, meaning there are components on both sides.

A. 10K SMD Resistors

- Solder 6 10K SMD resistor R01, R02, R03, R04, R06, R07.

B. 220K SMD Resistors

- Solder 5 10K SMD resistor R08, R09, R10, R11, R12.

C. Green SMD LED

- Solder a green SMD LED, LED01. Note the notch on the LED should point to the left.

D. 11-Pin Pocket Computer Connector

- Solder the 11-pin pocket computer connector. Be sure to solder this connector on the correct side of the PCB. It is to be soldered so that the pins come out on the side with the 1 LED and resistors, meaning the soldering happens on the same side as connector J2. *It is recommended to solder just 1 pin first, then ensure the connector pins are "level", adjusting the one soldered pin as needed. Once the connector is level, then solder the rest of the pins.*

E. 16-Pin IDC Connector

- Solder a 16-pin IDC connector J2. Be sure to solder this connector on the correct side of the PCB. It is to be soldered on the side without the resistors and 1 LED. **Note the notch orientation points downward**.